

## TECHNICAL STUDY COMPANION

# DiT 编解码器发展、架构、训练与数据工程

A Deep Insight into Encoders, Decoders and Visual Tokenizers for Diffusion Transformers

## DiT 编解码器技术演进 / Encoder-Decoder Evolution

从概率潜空间到高压压缩因果视频 tokenizer，再到面向复原的条件投影与 decoder-only 适配  
[pixels / frames] -> [continuous or discrete latent] -> [DiT tokens] -> [pixels / frames]

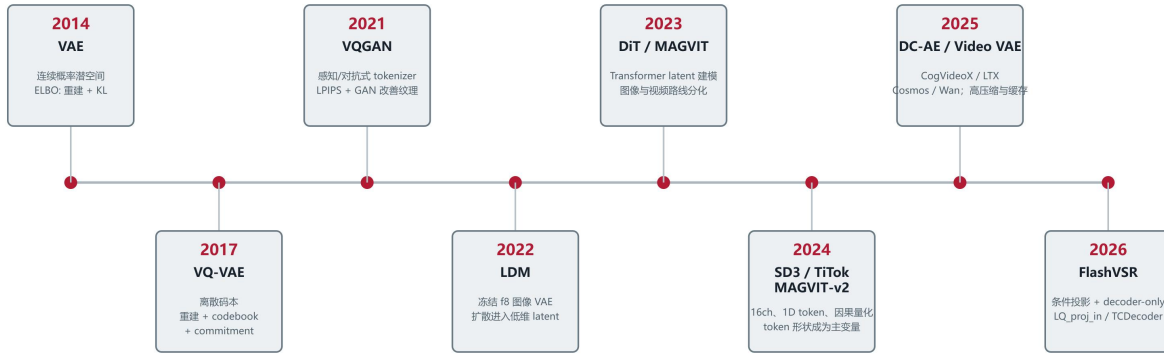


图 0-1 DiT 编解码器技术演进总览 / Encoder-Decoder Evolution Overview

证据锚点 / Evidence anchor: 解释图，依据本资料库架构矩阵与原始论文证据绘制。

中文主叙述 · English terminology · Paper-native figures · Code-level anchors

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摘要 / Abstract

编解码器在 DiT 系统中不是被动的输入输出接口，而是同时决定信息上限、token 数量、训练成本、时序边界和最终纹理风格。本报告从 VAE、VQ-VAE、VQGAN 和 LDM 的历史出发，系统比较 SD3、FLUX.1、TiTok、DC-AE，以及 MAGVIT、CogVideoX、Open-Sora、WF-VAE、VidTok、LTX-Video、Cosmos 与 Wan 等主流图像/视频编解码器；进一步把 tokenizer 预训练、冻结潜空间 DiT 训练和图像复原适配拆成三个不同优化问题，并以 FlashVSR 的 LQ\_proj\_in、WanDecoder 与 TCDecoder 为代码级案例。报告强调一个核心结论：压缩倍率、潜变量通道数、因果缓存和解码器监督共同构成生成模型的系统级设计空间，不能只以重建 PSNR 或单次推理速度孤立评价。

Encoders and decoders are not passive I/O components in a Diffusion Transformer system. They define the information ceiling, token budget, optimization difficulty, temporal boundary and final texture prior. This report traces the evolution from VAE, VQ-VAE, VQGAN and latent diffusion to modern image and video tokenizers, separates tokenizer pretraining from frozen-latent DiT training and restoration adaptation, and provides a code-level FlashVSR case study. The central systems insight is that compression ratio, latent dimensionality, causality, cache semantics and decoder supervision must be designed jointly.

**关键词 / Keywords** | Diffusion Transformer (DiT); Autoencoder; Variational Autoencoder (VAE); Visual Tokenizer; Causal Video VAE; Flow Matching; Perceptual Loss; FlashVSR; TCDecoder; Data Engineering.

证据口径与阅读方法

| 标签                     | 含义                       | 在本文中的使用方式             |
|------------------------|--------------------------|-----------------------|
| paper-verified         | 原论文正文、附录或原文图明确给出         | 可写作论文事实，并标注页码/图号      |
| code-verified          | 官方仓库源码或配置明确给出            | 可写作实现事实，并标注文件与行号      |
| primary-source-indexed | 权威原始来源已定位，但本地完整 PDF 尚未落盘 | 仅做基础定义或来源索引，不声称本地全文核验 |
| analysis               | 由多条证据归纳形成的工程解释           | 明确写为分析、建议或推论          |
| undisclosed            | 公开论文与源码未披露               | 保持空白边界，不用其他模型配方代填     |

**本地资料状态** | 来源索引共 40 条，当前有效完整论文 PDF 为 7 篇，官方源码快照 16/16 有效；另外

11 篇较新论文具有本地 arSiv 全文快照与原文图。VAE、VQ-VAE 等历史基础论文当前保留权威来源索引，完整 PDF 待补。

阅读导航

- 1. 先读第 1-2 章：建立 encoder、decoder、tokenizer、latent channel、compression factor 与 causality 的统一语义。
- 2. 再读第 3-4 章：分别理解图像与视频路线，重点观察 token 形状和 decoder 职责如何变化。
- 3. 第 5-6 章回答如何训练、LOSS 如何组合、数据如何制作和张量如何进入模型。
- 4. 第 7 章是 FlashVSR 深入案例，直接对应 LQ\_proj\_in、WanDecoder、TCDecoder 三个易混模块。
- 5. 第 8-9 章用于架构选型、复现检查和后续研究。Excel 是全量审计底稿，Word 是结构化学习手册。

1. 概念边界与历史演进

1.1 五个容易混淆的对象

| 对象    | 英文               | 核心职责                                    | 不应被误读为             |
|-------|------------------|-----------------------------------------|--------------------|
| 编码器   | Encoder          | 把像素/帧映射到潜变量；VAE 编码器还预测后验参数              | 任何把 RGB 投影到特征的条件分支 |
| 瓶颈    | Bottleneck       | 定义连续分布、离散码本或高压缩 token 形状                | 单纯的 DiT patchify   |
| 解码器   | Decoder          | 把潜变量恢复到像素/帧，可附带条件、去噪或缓存                 | 一定与编码器对称的固定逆运算     |
| 视觉分词器 | Visual Tokenizer | encoder-bottleneck-decoder 的完整 token 契约 | 只产生条件特征的 projector |
| 潜空间接口 | Latent Interface | 规定 DiT 输入的尺度、通道、归一化和解码方式                | 一个可随意替换而不影响主干的前处理  |

判断一个模块是不是 encoder，最实用的问题不是看名字，而是看它是否定义了完整的潜变量统计或 token 契约：输入是什么，输出能否被标准 decoder 接受，是否有 posterior/quantizer，是否与同一 tokenizer 的训练目标绑定。FlashVSR 的 LQ\_proj\_in 虽然把低清 RGB 压到与 Wan latent 对齐的尺度，但它不产生 Wan posterior，因此应称为条件投影器而不是 Wan encoder。

## 1.2 VAE：连续概率潜空间

VAE 的 encoder 学习近似后验  $q_{\phi}(z|x)$ ，decoder 学习  $p_{\theta}(x|z)$ 。重参数化  $z = \mu + \sigma * \epsilon$  让随机采样仍可反向传播。它建立了现代连续 latent autoencoder 的概率基础，但原始 VAE 并没有规定今天常见的 f8、4 通道或 16 通道接口。[1]

$$L_{VAE} = E_{q(z|x)}[-\log p_{\theta}(x|z)] + \beta * D_{KL}(q_{\phi}(z|x) || p(z))$$

*重建项 + KL 正则*

直觉上，重建项要求  $z$  保留输入信息；KL 项要求潜空间接近规则先验，便于生成模型采样。 $\beta$  过大时细节被压缩，过小时潜空间不规则。现代感知 VAE 通常加入 LPIPS 或 GAN，因此不能把原始 ELBO 等同于 SD/FLUX 类 autoencoder 的完整训练配方。

## 1.3 VQ-VAE：离散码本与 straight-through

VQ-VAE 把连续 encoder 特征映射到最近的 codebook entry，生成离散索引。量化不可导，因此使用 straight-through estimator 把 decoder 梯度近似传回 encoder；codebook loss 和 commitment loss 分别更新词典并约束 encoder 不要漂离码本。[2]

$$k^* = \operatorname{argmin}_k ||z_e(x) - e_k||_2 ; \quad z_q(x) = e_{k^*}$$

*nearest-neighbor vector quantization*

$$L_{VQ} = L_{rec} + ||sg[z_e] - e||_2^2 + \beta ||z_e - sg[e]||_2^2$$

*codebook + commitment*

**典型失败** | codebook collapse、dead codes 和过低的有效码本利用率会让名义 token 容量远大于实际容量。

## 1.4 VQGAN 与 LDM: 从压缩到感知重建

VQGAN 用感知损失和 patch discriminator 把重建目标从逐像素平均推进到局部纹理真实感；LDM 则把 autoencoder 与生成模型训练彻底分离：先训练 perceptual compression，再冻结 encoder/decoder，在低维 latent 上训练扩散模型。[4][5] 这一步奠定了今天 DiT 的主流系统边界。

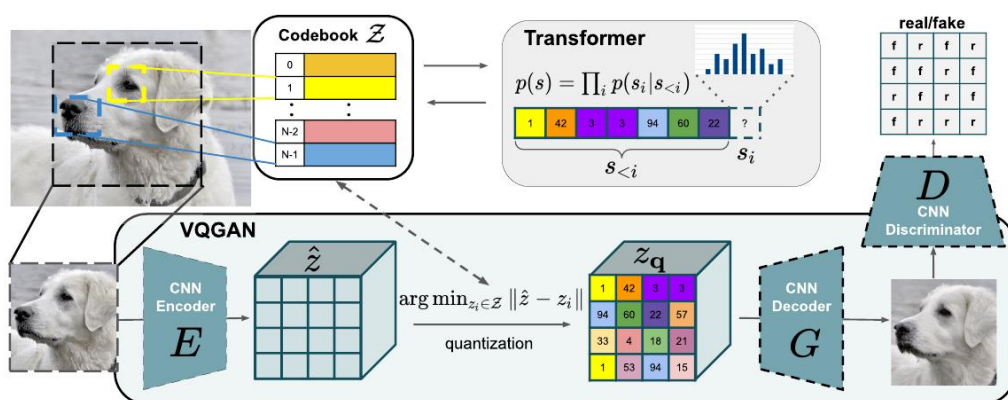


Figure 2. Our approach uses a convolutional VQGAN to learn a codebook of context-rich visual parts, whose composition is subsequently modeled with an autoregressive transformer architecture. A discrete codebook provides the interface between these architectures and a patch-based discriminator enables strong compression while retaining high perceptual quality. This method introduces the efficiency of convolutional approaches to transformer based high resolution image synthesis.

learns a GAN [19] on representations of an autoencoder and [20] on low-resolution wavelet coefficients which are then decoded to images with a learned generator.

[63] presents the Vector Quantised Variational Autoencoder (VQVAE), an approach to learn discrete representations of images, and models their distribution autoregressively with a convolutional architecture. [53] extends this approach to use a hierarchy of learned representations. However, these methods still rely on convolutional density

to higher resolutions.

High-resolution image synthesis requires a model that understands the global composition of images, enabling it to generate locally realistic as well as globally consistent patterns. Therefore, instead of representing an image with pixels, we represent it as a composition of perceptually rich image constituents from a codebook. By learning an effective code, as described in Sec. 3.1, we can significantly reduce the description length of compositions, which allows us to

图 1-1 VQGAN 的感知式离散 tokenizer 架构。

证据锚点 / Evidence anchor: VQGAN 原论文第 3 页 Fig.2; 本地 PDF: papers/01\_history\_and\_tokenizers/vqgan\_2012.09841.pdf。

so that producing 50k samples takes approximately 5 days [15] on a single A100 GPU. This has two consequences for the research community and users in general: Firstly, training such a model requires massive computational resources only available to a small fraction of the field, and leaves a huge carbon footprint [63, 83]. Secondly, evaluating an already trained model is also expensive in time and memory, since the same model architecture must run sequentially for a large number of steps (e.g. 25 - 1000 steps in [15]).

To increase the accessibility of this powerful model class and at the same time reduce its significant resource consumption, a method is needed that reduces the computational complexity for both training and sampling. Reducing the computational demands of DMs without impairing their performance is, therefore, key to enhance their accessibility.

**Departure to Latent Space** Our approach starts with the analysis of already trained diffusion models in pixel space: Fig. 2 shows the rate-distortion trade-off of a trained model. As with any likelihood-based model, learning can be roughly divided into two stages: First is a *perceptual compression* stage which removes high-frequency details but still learns little semantic variation. In the second stage, the actual generative model learns the semantic and conceptual composition of the data (*semantic compression*). We thus aim to first find a *perceptually equivalent, but computationally more suitable space*, in which we will train diffusion models for high-resolution image synthesis.

Following common practice [11, 23, 64, 65, 93], we separate training into two distinct phases: First, we train an autoencoder which provides a lower-dimensional (and thereby efficient) representational space which is perceptually equivalent to the data space. Importantly, and in contrast to previous work [23, 64], we do not need to rely on excessive spatial compression, as we train DMs in the learned

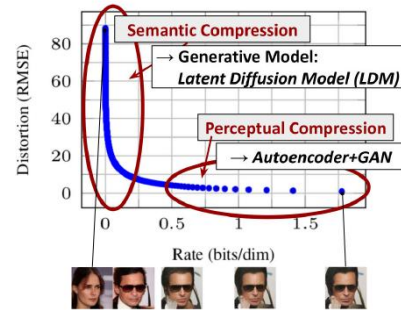


Figure 2. Illustrating perceptual and semantic compression: Most bits of a digital image correspond to imperceptible details. While DMs allow to suppress this semantically meaningless information by minimizing the responsible loss term, gradients (during training) and the neural network backbone (training and inference) still need to be evaluated on all pixels, leading to superfluous computations and unnecessarily expensive optimization and inference. We propose *latent diffusion models (LDMs)* as an effective generative model and a separate mild compression stage that only eliminates imperceptible details. Data and images from [29].

applied to high-resolution synthesis of megapixel images.

(ii) We achieve competitive performance on multiple tasks (unconditional image synthesis, inpainting, stochastic super-resolution) and datasets while significantly lowering computational costs. Compared to pixel-based diffusion approaches, we also significantly decrease inference costs.

(iii) We show that, in contrast to previous work [90] which learns both an encoder/decoder architecture and a score-based prior simultaneously, our approach does not require a delicate weighting of reconstruction and generative abilities. This ensures extremely faithful reconstructions

图 1-2 LDM 将感知压缩与 latent diffusion 解耦。

证据锚点 / Evidence anchor: LDM 原论文第 2 页; 本地 PDF: papers/01\_history\_and\_tokenizers/lDM\_2112.10752.pdf.

**系统结论** | 一旦 tokenizer 冻结, DiT 训练无法修复 decoder 的色偏、纹理涂抹或高频缺失。decoder 成为生成质量的硬上限, 同时 encoder 决定 DiT 看到的任务难度。

## 2. DiT 的潜空间接口与张量流

DiT 并不直接规定 encoder/decoder。经典 DiT 使用预训练 Stable Diffusion VAE, 把 RGB 图像压成 f8、4 通道 latent, 再按 p x p latent patch 转为 Transformer tokens。[6] 因此总 token 数由两次压缩共同决定: VAE 的空间压缩 f 和 DiT 的 patch size p。

$$N\_tokens(image) = (H / f / p) * (W / f / p)$$



图像 token 数

$$N\_tokens(video) = \text{ceil}(T / f\_t) * (H / f\_s / p\_h) * (W / f\_s / p\_w)$$

视频 token 数

## 主流 DiT 编解码器张量流 / Tensor Flow

图像与视频使用同一训练边界：先训练 tokenizer，再冻结 latent 接口训练 DiT；复原模型可额外注入观测条件。

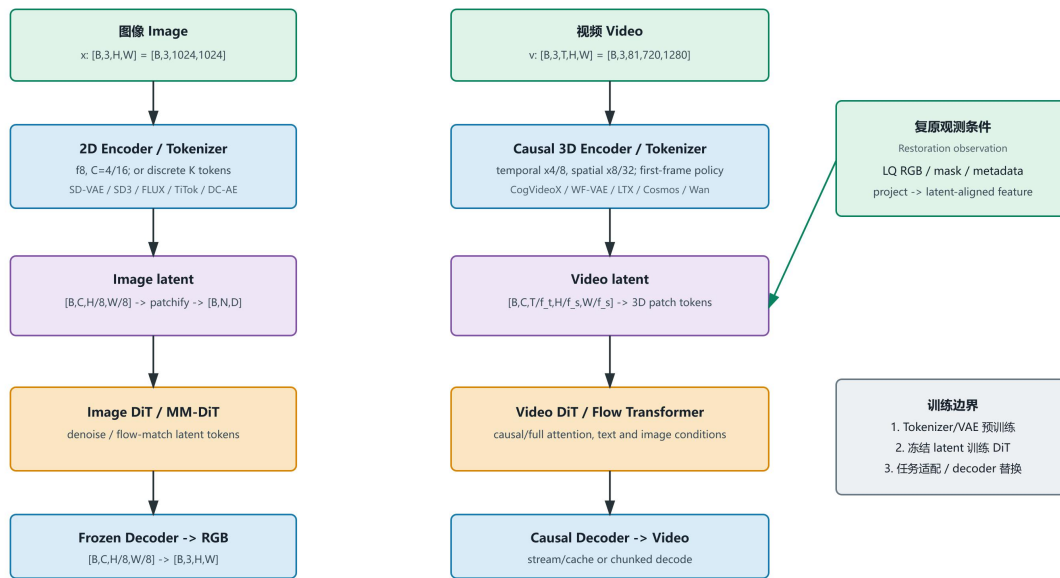


图2-1 图像/视频tokenizer与DiT的张量流。

证据锚点 / Evidence anchor: 解释图；数值来自 DiT、SD3、CogVideoX、LTX-Video、Wan 与 FlashVSR 的公开结构。

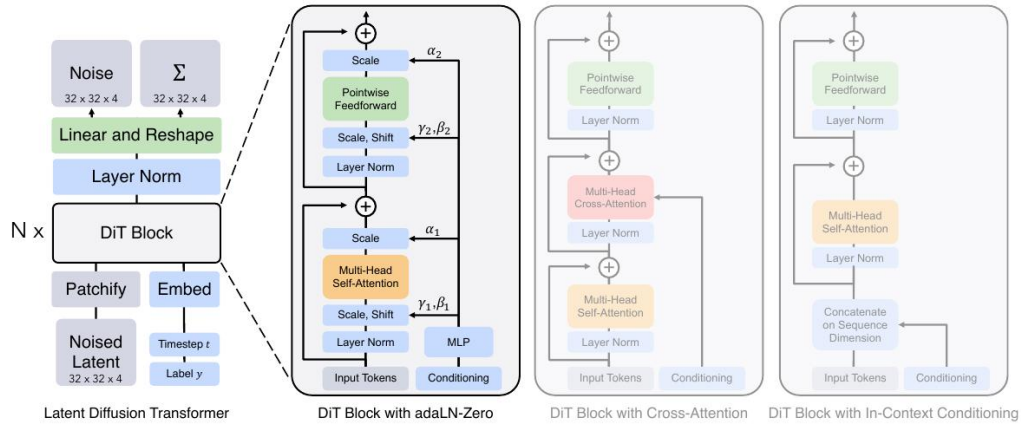


Figure 3. **The Diffusion Transformer (DiT) architecture.** *Left:* We train conditional latent DiT models. The input latent is decomposed into patches and processed by several DiT blocks. *Right:* Details of our DiT blocks. We experiment with variants of standard transformer blocks that incorporate conditioning via adaptive layer norm, cross-attention and extra input tokens. Adaptive layer norm works best.

**Denosing diffusion probabilistic models (DDPMs).** Diffusion [19, 54] and score-based generative models [22, 56] have been particularly successful as generative models of images [35, 46, 48, 50], in many cases outperforming generative adversarial networks (GANs) [12] which had previously been state-of-the-art. Improvements in DDPMs over the past two years have largely been driven by improved sampling techniques [19, 27, 55], most notably classifier-free guidance [21], reformulating diffusion models to predict noise instead of pixels [19] and using cascaded DDPM pipelines where low-resolution base diffusion models are trained in parallel with upsamplers [9, 20]. For all the diffusion models listed above, convolutional U-Nets [49] are the de-facto choice of backbone architecture. Concurrent work [24] introduced a novel, efficient architecture based on attention for DDPMs; we explore pure transformers.

### 3. Diffusion Transformers

#### 3.1. Preliminaries

**Diffusion formulation.** Before introducing our architecture, we briefly review some basic concepts needed to understand diffusion models (DDPMs) [19, 54]. Gaussian diffusion models assume a forward noising process which gradually applies noise to real data  $x_0$ :  $q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_0, (1 - \alpha_t)\mathbf{I})$ , where constants  $\alpha_t$  are hyperparameters. By applying the reparameterization trick, we can sample  $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon_t$ , where  $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$ .

Diffusion models are trained to learn the reverse process that inverts forward process corruptions:  $p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t), \Sigma_\theta(x_t))$ , where neural networks are used to predict the statistics of  $p_\theta$ . The reverse process model is

图2-2 DiT 原论文: latent patchify、Transformer blocks 与输出头。

证据锚点 / Evidence anchor: DiT 原论文第3 页 Fig.3; [source\\_code/dit/models.py](#).

**形状示例** | 512x512 RGB 经 f8 VAE 得到 [B,4,64,64]。若 DiT patch size  $p=2$ ，则 token grid 为 32x32，共 1024 tokens；若改成 16 通道 latent，token 数不变，但每个位置的信息维度和预测难度显著增加。



### 3. 主流图像编解码器

#### 3.1 SD3 VAE 与 MM-DiT：16 通道 f8 接口

SD3 将图像 latent 从经典 4 通道扩展到 16 通道，同时保持 f8 空间压缩。官方参考实现使用四级  $ch\_mult=(1,2,4,4)$ 、两层 residual blocks 和 mid attention，encoder 输出  $2 * z\_channels$  以形成后验参数；latent 还需经过  $scale=1.5305$ 、 $shift=0.0609$  的归一化接口。[8] 更高通道容量改善重建，但也扩大 flow model 的输入/输出维度。

| 结构锚点             | SD3 实现                                                     | 对 MM-DiT 的影响                                    |
|------------------|------------------------------------------------------------|-------------------------------------------------|
| Encoder backbone | 2D ResNet; $ch\_mult=(1,2,4,4)$ ; 每级 2 个 residual blocks   | 保持 f8 网格，同时扩大单个 latent 位置的表达能力                  |
| Bottleneck       | mid-block self-attention                                   | 在进入 flow model 前补充低分辨率全局信息交互                    |
| Posterior        | encoder 输出 $2 \times 16$ channels, 拆分为 mean / log-variance | 采样后以 16-channel continuous latent 作为图像 token 来源 |
| Latent contract  | $scale=1.5305$ ; $shift=0.0609$                            | MM-DiT 权重依赖该统计接口，不可与其他 VAE latent 直接互换          |

Scaling Rectified Flow Transformers for High-Resolution Image Synthesis

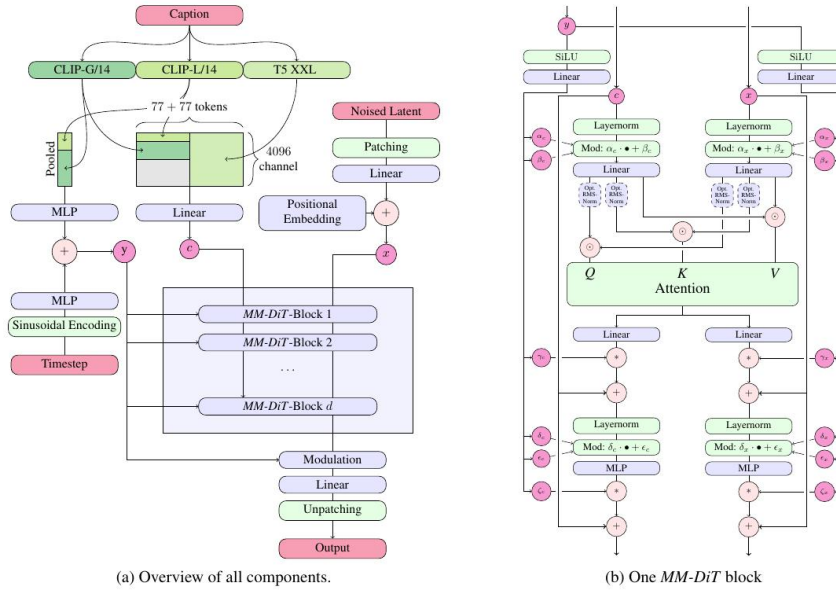


Figure 2. **Our model architecture.** Concatenation is indicated by  $\odot$  and element-wise multiplication by  $*$ . The RMS-Norm for  $Q$  and  $K$  can be added to stabilize training runs. Best viewed zoomed in.

concatenate the two sequences. We then follow DiT and apply a sequence of modulated attention and MLPs.

Since text and image embeddings are conceptually quite different, we use two separate sets of weights for the two modalities. As shown in Figure 2b, this is equivalent to having two independent transformers for each modality, but joining the sequences of the two modalities for the attention operation, such that both representations can work in their own space yet take the other one into account.

For our scaling experiments, we parameterize the size of the model in terms of the model’s depth  $d$ , i.e. the number of attention blocks, by setting the hidden size to  $64 \cdot d$  (expanded to  $4 \cdot 64 \cdot d$  channels in the MLP blocks), and the number of attention heads equal to  $d$ .

## 5. Experiments

### 5.1. Improving Rectified Flows

In addition, the losses of different approaches are incomparable and also do not necessarily correlate with the quality of output samples; hence we need evaluation metrics that allow for a comparison between approaches. We train models on ImageNet (Russakovsky et al., 2014) and CC12M (Changpinyo et al., 2021), and evaluate both the training and the EMA weights of the models during training using validation losses, CLIP scores (Radford et al., 2021; Hessel et al., 2021), and FID (Heusel et al., 2017) under different sampler settings (different guidance scales and sampling steps). We calculate the FID on CLIP features as proposed by (Sauer et al., 2021). All metrics are evaluated on the COCO-2014 validation split (Lin et al., 2014). Full details on the training and sampling hyperparameters are provided in Appendix B.3.

#### 5.1.1. RESULTS

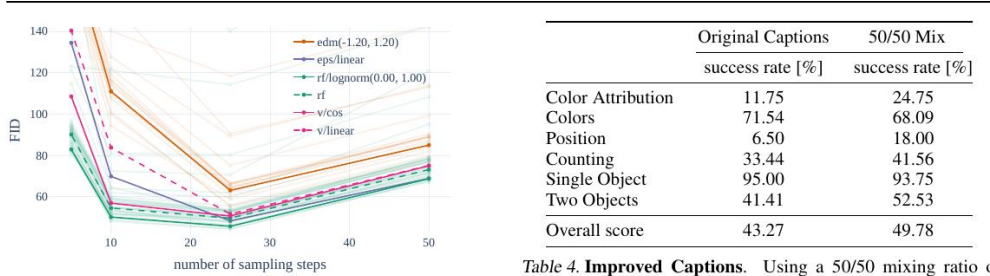
We train each of 61 different formulations on the two datasets. We include the following variants from Section 3:

- Both  $\epsilon_q$  and  $\epsilon_v$ -prediction, loss with linear

图 3-1 SD3 的 MM-DiT 双模态联合注意力结构。

证据锚点 / Evidence anchor: SD3 原论文第 5 页; 本地 PDF: papers/02\_image\_vae/sd3\_2403.03206.pdf。

Scaling Rectified Flow Transformers for High-Resolution Image Synthesis



**Figure 3. Rectified flows are sample efficient.** Rectified Flows perform better than other formulations when sampling fewer steps. For 25 and more steps, only  $rf/lognorm(0.00, 1.00)$  remains competitive to  $eps/linear$ .

| Metric                                 | 4 chn | 8 chn | 16 chn       |
|----------------------------------------|-------|-------|--------------|
| FID ( $\downarrow$ )                   | 2.41  | 1.56  | <b>1.06</b>  |
| Perceptual Similarity ( $\downarrow$ ) | 0.85  | 0.68  | <b>0.45</b>  |
| SSIM ( $\uparrow$ )                    | 0.75  | 0.79  | <b>0.86</b>  |
| PSNR ( $\uparrow$ )                    | 25.12 | 26.40 | <b>28.62</b> |

**Table 3. Improved Autoencoders.** Reconstruction performance metrics for different channel configurations. The downsampling factor for all models is  $f = 8$ .

cordingly, the final performance of our algorithm depends not only on the training formulation, but also on the parameterization via a neural network and the quality of the image and text representations we use. In the following sections, we describe how we improve all these components before scaling our final method in Section 5.3.

#### 5.2.1. IMPROVED AUTOENCODERS

Latent diffusion models achieve high efficiency by operating in the latent space of a pretrained autoencoder (Rombach et al., 2022), which maps an input RGB  $X \in \mathbb{R}^{H \times W \times 3}$  into a lower-dimensional space  $x = E(X) \in \mathbb{R}^{h \times w \times d}$ . The reconstruction quality of this autoencoder provides an upper

|                   | Original Captions | 50/50 Mix        |
|-------------------|-------------------|------------------|
|                   | success rate [%]  | success rate [%] |
| Color Attribution | 11.75             | 24.75            |
| Colors            | 71.54             | 68.09            |
| Position          | 6.50              | 18.00            |
| Counting          | 33.44             | 41.56            |
| Single Object     | 95.00             | 93.75            |
| Two Objects       | 41.41             | 52.53            |
| Overall score     | 43.27             | 49.78            |

**Table 4. Improved Captions.** Using a 50/50 mixing ratio of synthetic (via CogVLM (Wang et al., 2023)) and original captions improves text-to-image performance. Assessed via the GenEval (Ghosh et al., 2023) benchmark.

nature of the human-generated captions that come with large-scale image datasets, which overly focus on the image subject and usually omit details describing the background or composition of the scene, or, if applicable, displayed text (Betker et al., 2023). We follow their approach and use an off-the-shelf, state-of-the-art vision-language model, CogVLM (Wang et al., 2023), to create synthetic annotations for our large-scale image dataset. As synthetic captions may cause a text-to-image model to forget about certain concepts not present in the VLM’s knowledge corpus, we use a ratio of 50 % original and 50 % synthetic captions.

To assess the effect of training on this caption mix, we train two  $d = 15$  MM-DiT models for 250k steps, one on only original captions and the other on the 50/50 mix. We evaluate the trained models using the GenEval benchmark (Ghosh et al., 2023) in Table 4. The results demonstrate that the model trained with the addition of synthetic captions clearly outperforms the model that only utilizes original captions. We thus use the 50/50 synthetic/original caption mix for the remainder of this work.

#### 5.2.3. IMPROVED TEXT-TO-IMAGE BACKBONES

图3-2 SD3 原论文的autoencoder 架构比较与16 通道选择。

证据锚点 / Evidence anchor: SD3 原论文第7 页 Table 3; source\_code/sd3/sd3\_impls.py:97-107,247-310。

## 3.2 FLUX.1 AE：公开结构与未公开训练配方

FLUX.1 官方源码公开了与 SD3 同属 2D ResNet autoencoder 家族的结构：in/out RGB=3, base channel=128, ch\_mult=[1,2,4,4], 两层 residual blocks, z\_channels=16, 四个 resolution levels 对应 f8; latent 归一化为  $z\_norm = 0.3611 * (z - 0.1159)$ 。源码可核验 encoder、decoder 与归一化，但没有公开完整 AE 训练数据、损失权重、判别器策略与优化日程。

$$z\_model = 0.3611 * (z\_ae - 0.1159) ; \quad z\_ae = z\_model / 0.3611 + 0.1159$$

FLUX.1 latent normalization

| 维度              | SD3                         | FLUX.1                      | 工程含义                  |
|-----------------|-----------------------------|-----------------------------|-----------------------|
| 空间压缩            | f8                          | f8（由源码层级推得）                 | token 网格规模相近          |
| latent channels | 16                          | 16                          | 比经典 SD-VAE 的 4ch 更高容量 |
| 归一化             | scale 1.5305 / shift 0.0609 | scale 0.3611 / shift 0.1159 | 权重与 latent 不可交叉混用     |
| AE 训练配方         | 论文未完整披露                     | 未公开                         | 只能复用架构，不能声称复现官方训练     |

证据锚点 / Evidence anchor: `source_code/flux/modules/autoencoder.py`; `source_code/flux/util.py:539-548`; 证据状态 `code-verified+undisclosed`。

**关键边界** | 把 LDM 或 SD3 的 LPIPS/GAN/KL 权重复制给 FLUX autoencoder，可以作为复现实验起点，但必须标注为 engineering baseline，不能写成 FLUX 官方事实。

3.3 TiTok：从二维网格到一维少量 token

TiTok 不再把 bottleneck 固定成二维格点，而是让 ViT encoder 同时处理图像 patches 与 K 个可学习 latent tokens，再量化为 K 个一维 tokens。旗舰设置在 256px 使用 32 tokens、codebook 4096。它主要服务 MaskGIT 路线，但对 DiT 的启示非常直接：token 预算不一定只能依靠更大空间压缩获得，也可以改变 token 拓扑。[9]

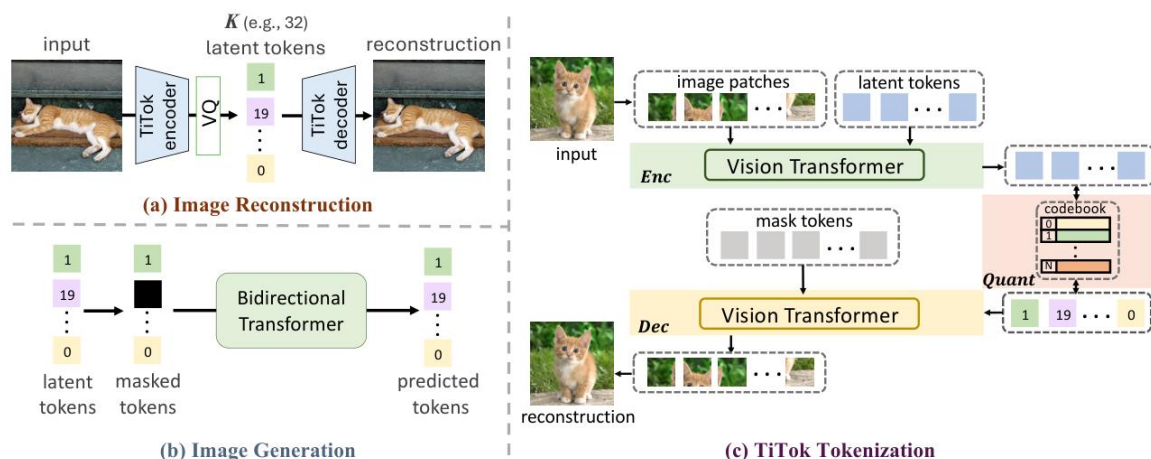


图3-3 TiTok:  $K$  个一维 latent tokens 的编码与解码框架。

证据锚点 / Evidence anchor: TiTok 原文 ar5iv 快照 Fig.3; text/ar5iv\_2406.07550.txt。

训练采用两阶段策略：先预测 MaskGIT-VQGAN 的 proxy codes，降低直接 RGB/GAN 优化难度；再冻结 encoder 与 quantizer，仅 fine-tune decoder，使用 VQGAN 风格的像素、感知与对抗目标。该方法说明 decoder 可以被视为独立的高频重建器进行后训练。

### 3.4 DC-AE：高压压缩不是多堆几层 stride

DC-AE 把空间压缩推进到 f32/f64/f128，代表一种明显的系统转向：把 token 压缩从 DiT patchify 前移到 autoencoder。本质难点不是网络能否 downsample，而是如何在高压压缩下保持全局结构、局部细节和可优化性。DC-AE 使用 residual autoencoding、space-to-channel shortcut，以及低分辨率预训练、高分辨率 latent adaptation、局部 GAN refinement 的分阶段方案。[10]



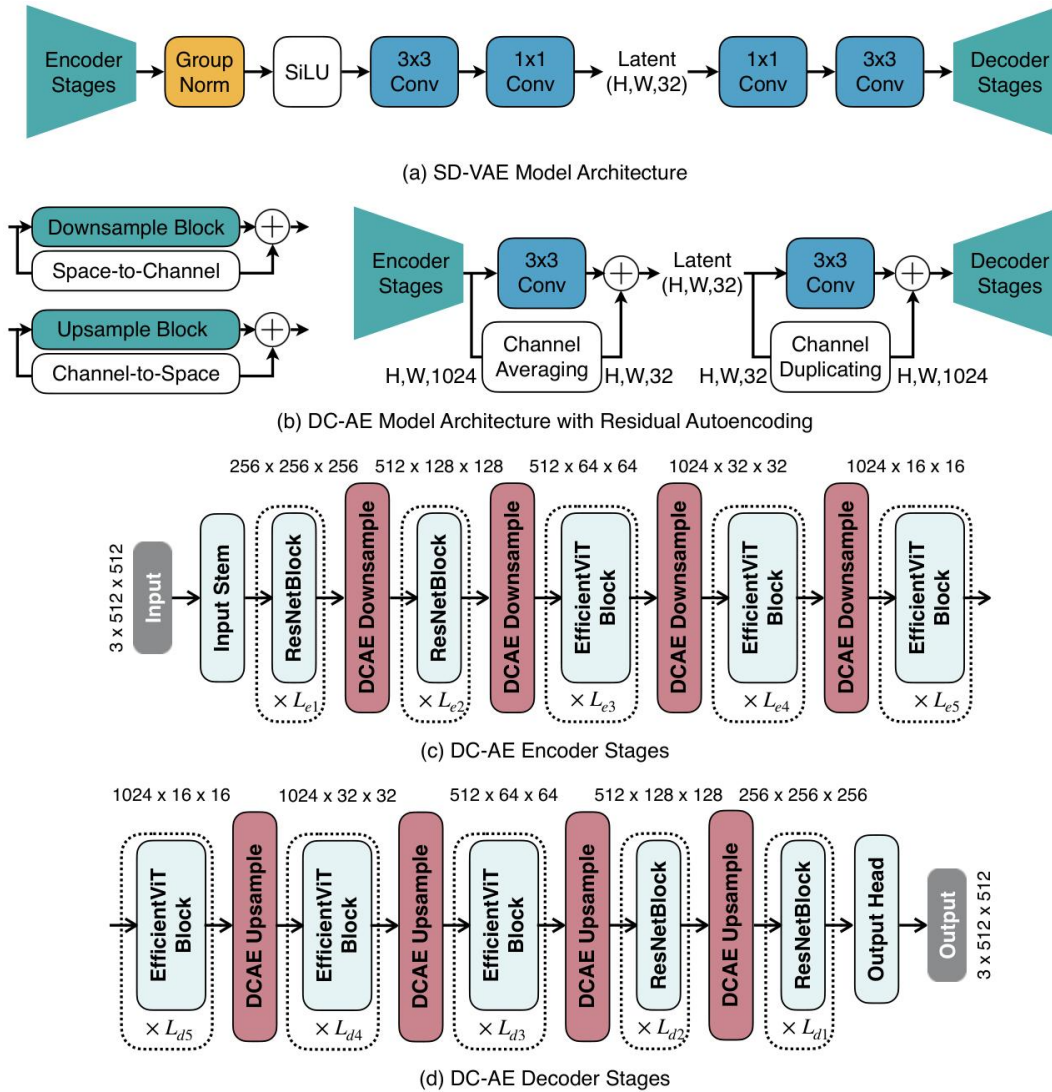


图3-4 DC-AE 的 residual autoencoding 与高压压缩结构。

证据锚点 / Evidence anchor: DC-AE 原文 ar5iv 资产; text/ar5iv\_2410.10733.txt。

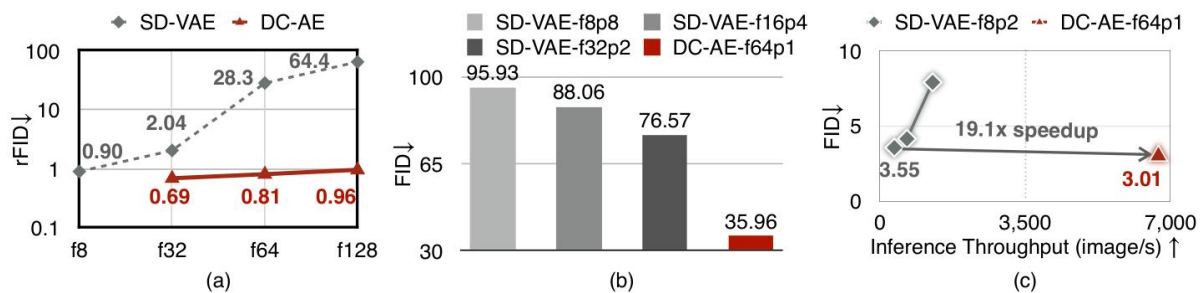


图3-5 DC-AE 展示不同压缩等级对重建与生成效率的影响。

证据锚点 / Evidence anchor: DC-AE 原文 Fig.1 资产。

**工程判断** | 高压压缩 VAE 适合高分辨率生成和长视频，但不天然适合严格复原。压缩越激进，decoder



越可能依靠生成先验补纹理，忠实度风险越高。

## 4. 主流视频编解码器

视频 tokenizer 必须额外处理时间压缩、第一帧策略、因果卷积、长视频缓存和帧间一致性。名义上相同的 4x8x8 压缩，如果 first-frame policy、normalization 或 cache contract 不同，也不能互换。

| 模型            | 类型            | 压缩              | 关键特征                                  |
|---------------|---------------|-----------------|---------------------------------------|
| MAGVIT / v2   | 离散            | 配置相关            | 3D token grid；v2 使用 causal conv 与 LFQ |
| CogVideoX     | 连续 causal VAE | 4x8x8           | context parallelism；Frame Pack；多通道配置  |
| Open-Sora 1.2 | 2D+3D hybrid  | 4x8x8           | 冻结 SDXL VAE；逐步从 feature 到 RGB 目标      |
| WF-VAE        | 连续 causal VAE | 常见视频倍率          | wavelet energy flow；causal cache      |
| LTX-Video     | denoising VAE | 8x32x32         | decoder 兼任最后去噪步骤                      |
| Cosmos        | 连续+离散         | 4/8x8x8；8x16x16 | 统一图像视频；可选 diffusion decoder           |
| Wan           | 连续 causal VAE | 4x8x8, 16ch     | 首帧+4 帧块；feature cache；FlashVSR 基础     |

### 4.1 MAGVIT 与 MAGVIT-v2：离散视频 token 的祖先路线

MAGVIT 通过 3D VQ tokenizer 统一图像与视频任务，为后续视频 token 建模提供重要基线。[13] MAGVIT-v2 进一步采用 temporally causal convolution 和 lookup-free quantization (LFQ)，用二值因子化方式扩展码本，减少传统 VQ lookup 的瓶颈。[14] 即使现代视频 DiT 多用连续 VAE，这条路线仍深刻影响 causal convolution、first-frame handling 和 tokenizer benchmark。

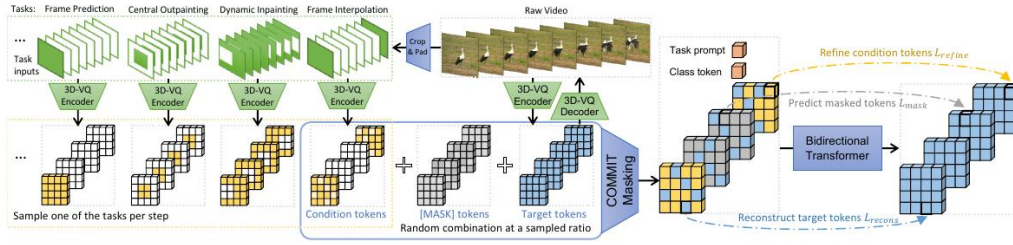


Figure 2. **MAGVIT pipeline overview.** The 3D-VQ encoder quantizes a video into discrete tokens, while the 3D-VQ decoder maps them back to the pixel space. We sample one of the tasks at each training step and build its condition inputs by cropping and padding the raw video, where green denotes valid pixels and white is padding. We quantize the condition inputs with the 3D-VQ encoder and select the non-padding part as condition tokens. The masked token sequence combines condition tokens, [MASK] tokens, and the target tokens, with a task prompt and a class token as the prefix. The bidirectional transformer learns to predict the target tokens through three objectives: refining condition tokens, predicting masked tokens, and reconstructing target tokens.

Fig. 2 illustrates the training in the second stage. At each training step, we sample one of the tasks with its prompt token, obtain a task-specific conditional mask, and optimize the transformer to predict all target tokens given masked inputs. During inference, we adapt the non-autoregressive decoding method to generate tokens conditionally on the task-specific inputs, which will be detailed in Algorithm 1.

### 3.1. Spatial-Temporal Tokenization

Our video VQ autoencoder is built upon the image VQ-GAN [20]. Let  $\mathbf{V} \in \mathbb{R}^{T \times H \times W \times 3}$  be a video clip of  $T$  frames. The VQ encoder tokenizes the video as  $f_T: \mathbf{V} \rightarrow \mathbf{z} \in \mathbb{Z}^N$ , where  $\mathbb{Z}$  is the codebook. The decoder  $f_T^{-1}$  maps the latent tokens back to video pixels.

The VQ autoencoder is a crucial module as it not only sets a quality bound for the generation but also determines the token sequence length, hence affecting generation efficiency. Existing methods apply VQ encoders either on each frame independently (2D-VQ) [26, 41] or on a super-

**Inflation and padding.** We initialize our 3D-VQ with weights from a 2D-VQ in a matching architecture to transfer learned spatial relationships [11], known as 3D inflation. We use inflation on small datasets such as UCF-101 [55]. We use a central inflation method for the convolution layers, where the corresponding 2D kernel fills in the temporally central slice of a zero-filled 3D kernel. The parameters of the other layers are directly copied. To improve token consistency for the same content at different locations [21], we replace the same (zero) padding in the convolution layers with reflect padding, which pads with non-zero values.

**Training.** We apply the image perceptual loss [20] on each frame. The LeCam regularization [58] is added to the GAN loss to improve the training stability. We adopt the discriminator architecture from StyleGAN [38] and inflate it to 3D. With these components, unlike VQGAN, our model is trained stably with GAN loss from the beginning.

### 3.2. Multi-Task Masked Token Modeling

图 4-1 MAGVIT 的统一视频tokenizer 与生成管线。

证据锚点 / Evidence anchor: MAGVIT 原论文第 3 页 Fig.2; 本地 PDF: papers/03\_video\_vae/magvit\_2212.05199.pdf。

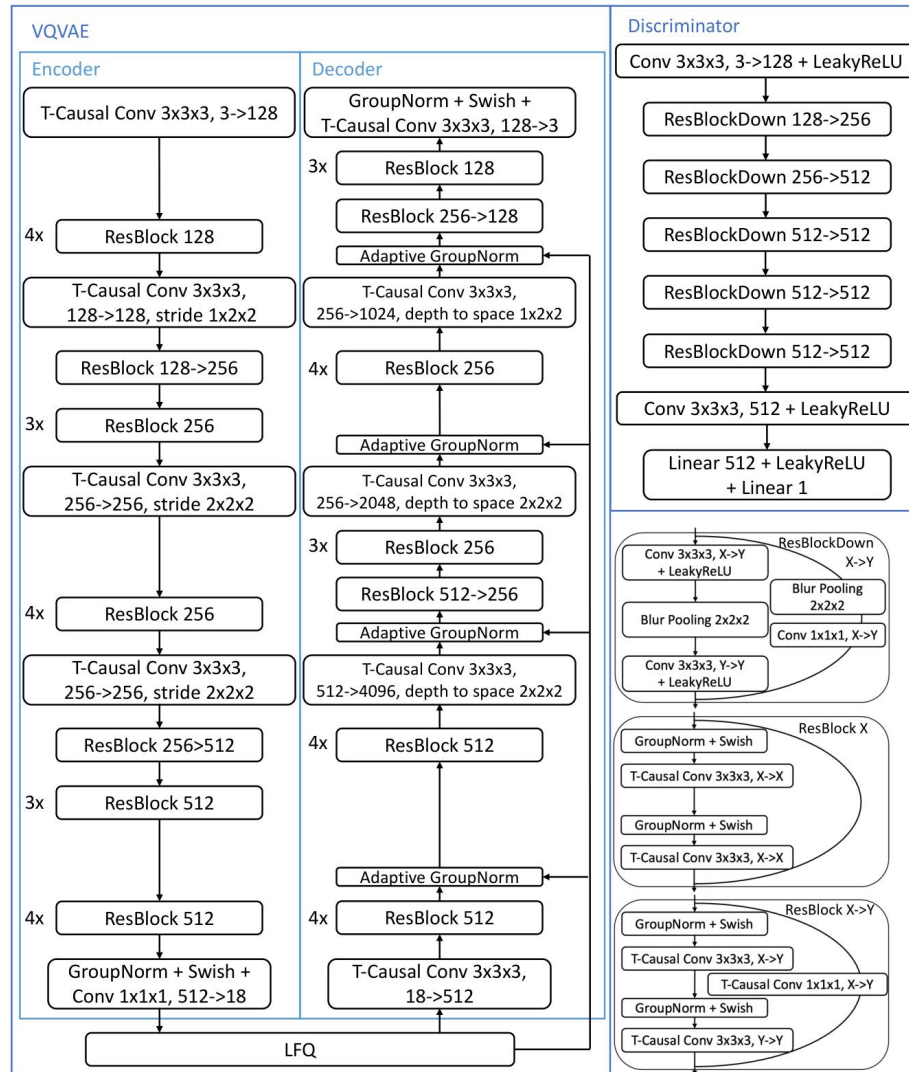


图4-2 MAGViT-v2 的causal tokenizer 与LFQ 结构。

证据锚点 / Evidence anchor: MAGViT-v2 原文 ar5iv 资产; text/ar5iv\_2310.05737.txt.

## 4.2 CogVideoX 3D VAE: 生产级 4x8x8 causal latent

CogVideoX 采用 temporally causal 3D convolutional VAE, 把时间、空间分别压缩 4、8、8 倍。论文同时研究 4/8/16/32 latent channels, 提醒复现者不能只凭模型家族名假设所有 checkpoint 的通道数一致。官方实现还引入 context parallelism, 解决大视频编码解码的显存瓶颈。[16]

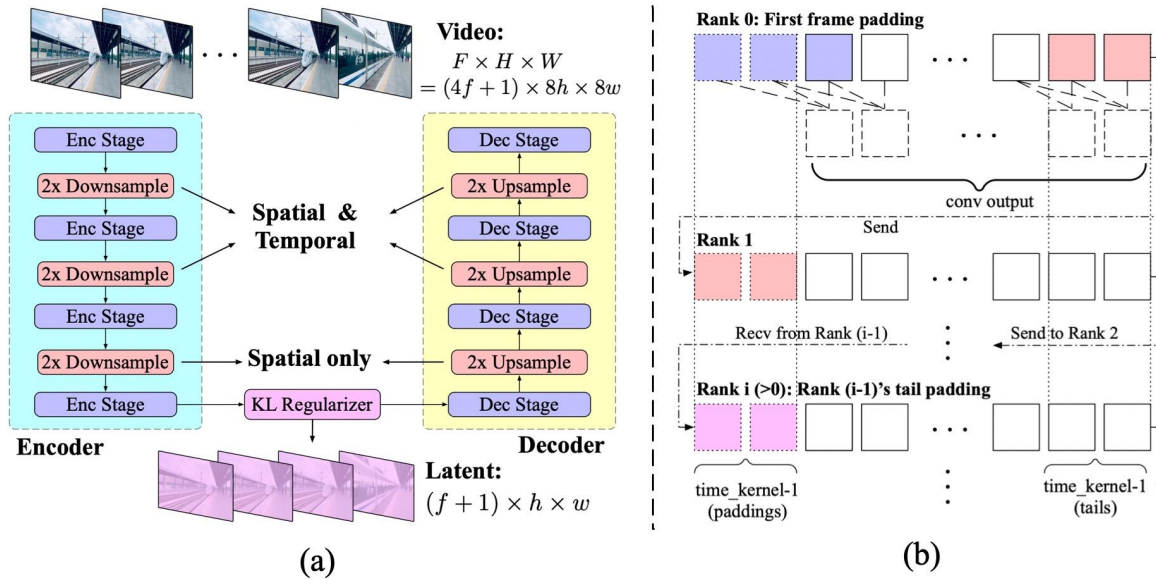


图4-3 CogVideoX 3D causal VAE 与压缩路径。

证据锚点 / Evidence anchor: CogVideoX 原文 Fig.4; source\_code/cogvideox/sat/vae\_modules/cp\_enc\_dec.py:743-998。

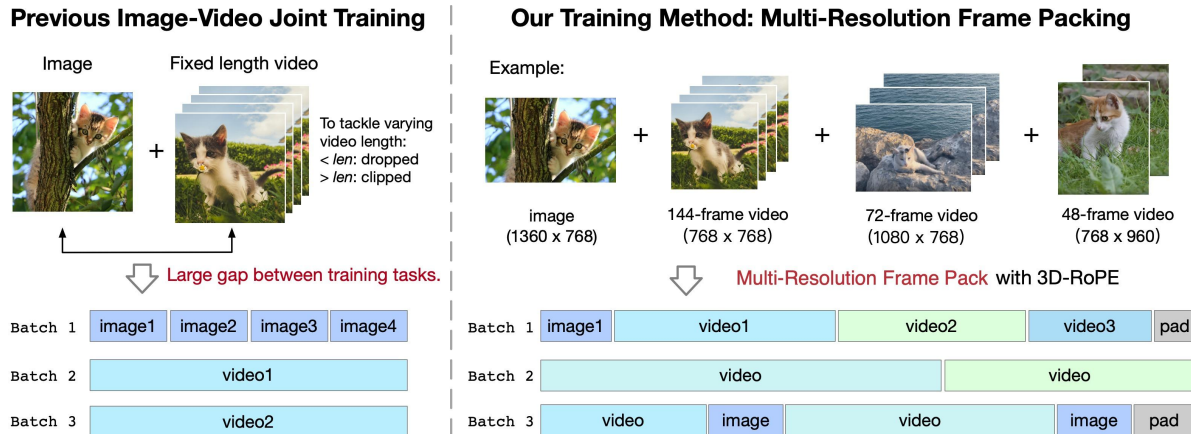


图4-4 CogVideoX Frame Pack: 混合时长训练的batch 组织方式。

证据锚点 / Evidence anchor: CogVideoX 原文图; text/ar5iv\_2408.06072.txt。

数据侧，CogVideoX 把 caption density 当作模型质量杠杆，使用多阶段质量与 caption pipeline，并在末期使用高质量 cooling subset。VAE 的 self-reconstruction 数据与 DiT 的 video-caption 数据相关但目标不同，应分别管理。

### 4.3 Open-Sora 1.2: 用冻结图像 VAE 启动视频 VAE

Open-Sora 1.2 先冻结 SDXL 2D VAE，再叠加 MAGVIT-v2 风格的 3D VAE。训练不是一次性从 RGB 到 RGB，而是三阶段改变 target space：先重建 2D-VAE features 并做 identity alignment，再强化 feature reconstruction，最后直接重建 RGB 视频。其公开日程为 0-380K、380K-640K、640K-1.2M steps。[19]

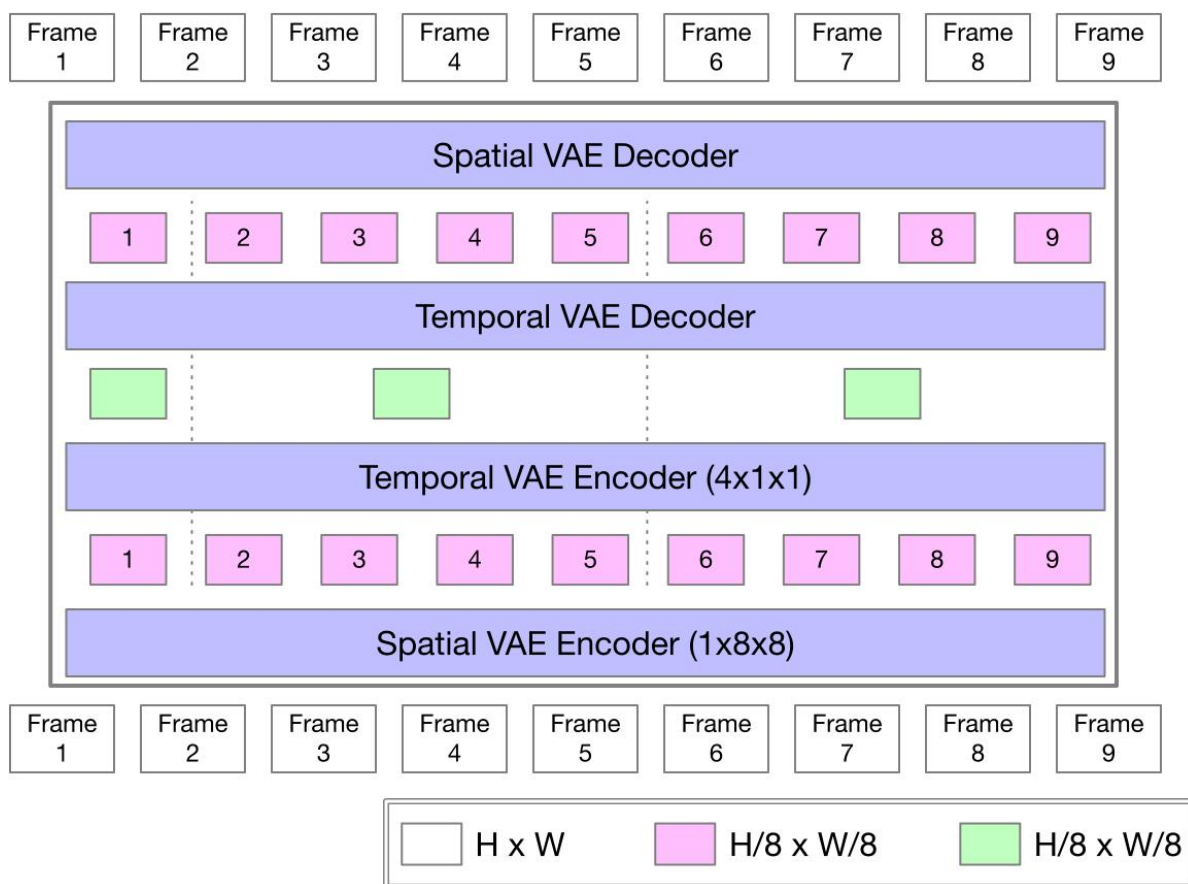


图4-5 Open-Sora 1.2 的2D VAE + 3D VAE 混合结构。

证据锚点 / Evidence anchor: Open-Sora 原文 Fig.5 的 ar5iv SVG 渲染; text/ar5iv\_2412.20404.txt。

**训练启示 |** 当视频 VAE 难以从头稳定训练时，可以先对齐成熟图像 VAE 的 feature space，再逐步切换到直接 RGB 视频监督。



#### 4.4 WF-VAE：小波能量流与 block-wise 解码

WF-VAE 在 causal 3D backbone 之外加入 multi-level wavelet energy-flow path，让低频结构和高频细节以显式能量流进入编码/解码；同时用 causal cache 支持分块推理。论文消融中 wavelet loss 权重  $\lambda_{WL}=0.1$ ，且 L1 在其设置下优于 L2。[17]

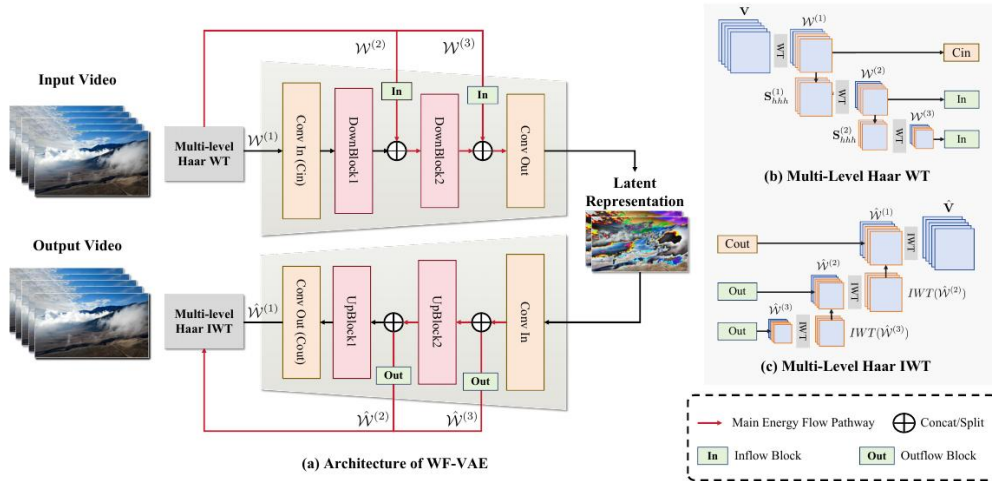


Figure 2. **Overview of WF-VAE.** Our architecture consists of a backbone and a main energy flow pathway. The pathway functions as a “highway” for the main flow of video energy, channeling this energy into the backbone through concatenations, allowing more critical video information to be preserved in the latent representation.

$$\mathbf{S}_{ijk}^{(l)} = \mathbf{S}^{(l-1)} * (f_i \otimes f_j \otimes f_k), \quad (1)$$

where  $f_i, f_j, f_k \in \{\mathbf{h}, \mathbf{g}\}$  represent the filters applied along each dimension, and  $*$  represents the convolution operation. The transform begins with  $\mathbf{S}^{(0)} = \mathbf{V}$ , and for subsequent layers,  $\mathbf{S}^{(l)} = \mathbf{S}_{hhh}^{(l-1)}$ , indicating that each layer operates on the low-frequency component from the previous layer. At each decomposition layer  $l$ , the transform produces eight sub-band components:  $\mathcal{W}^{(l)} = \{\mathbf{S}_{hhh}^{(l)}, \mathbf{S}_{hhg}^{(l)}, \mathbf{S}_{hgh}^{(l)}, \mathbf{S}_{ghh}^{(l)}, \mathbf{S}_{hgg}^{(l)}, \mathbf{S}_{ggh}^{(l)}, \mathbf{S}_{ghg}^{(l)}, \mathbf{S}_{ggg}^{(l)}\}$ . Here,  $\mathbf{S}_{hhh}^{(l)}$  represents the low-frequency component across all

rates to high-frequency details. With the additional path, we can reduce the computational cost brought by the dense 3D convolutions in the backbone.

Specifically, given a video  $\mathbf{V}$ , we apply multi-level wavelet transform to obtain pyramid features  $\mathcal{W}^{(1)}, \mathcal{W}^{(2)}$  and  $\mathcal{W}^{(3)}$ . We utilize  $\mathcal{W}^{(1)}$  as the input for the encoder and the target output for the decoder. The backbone and multi-level wavelet transform downsample the feature maps simultaneously at every downsampling layer, enabling the concatenation of features from two branches in the backbone. We employ Inflow Block to transform the channel numbers of  $\mathcal{W}^{(2)}$  and  $\mathcal{W}^{(3)}$  to  $C_{in}$ , which are then con-

图4-6 WF-VAE 的 wavelet-driven energy flow 架构。

证据锚点 / Evidence anchor: WF-VAE 原论文第3页Fig.2; 本地PDF: papers/03\_video\_vae/wfvae\_2411.17459.pdf。

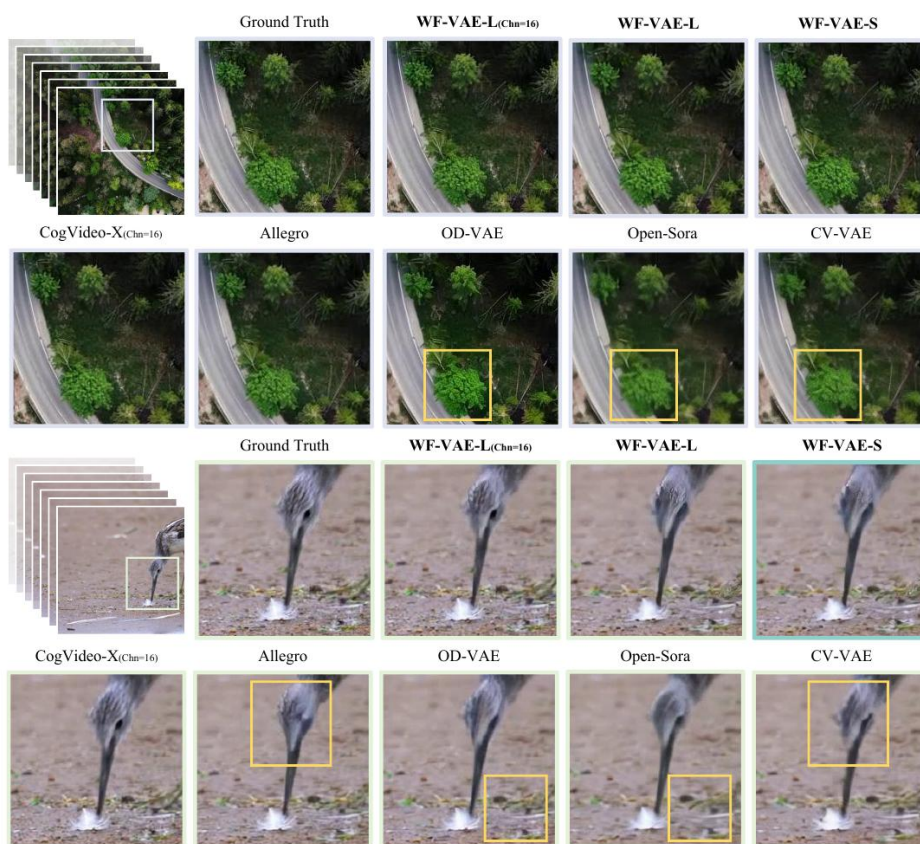


Figure 5. **Qualitative comparison of reconstruction performance.** We select two scenarios to comprehensively evaluate the visual quality of videos reconstructed by existing VAEs. Top: scenario contains **rich details**. Bottom: scenario contains **fast motion**.

ing and data processing. In terms of encoding speed, WF-VAE-L achieves an encoding time of 0.0513 seconds, while OD-VAE, CogVideoX, and Allegro VAE exhibit encoding times of 0.0945 seconds, 0.1810 seconds, and 0.3731 sec-

complexity, WF-VAE-L competes well with Allegro [50], outperforming it in PSNR, LPIPS and FVD but slightly lagging in SSIM. However, WF-VAE-L is significantly more computationally efficient than Allegro. Additionally, we

图 4-7 WF-VAE 原文重建效果对比。

证据锚点 / Evidence anchor: WF-VAE 原论文第 6 页; 同上本地 PDF。

## 4.5 VidTok: 把训练变量变成可控实验

VidTok 提供 continuous KL 与 discrete FSQ 等多种 bottleneck, 用统一协议研究架构、量化、帧率和分辨率。其重要结论之一是: 先低分辨率训练完整 tokenizer, 再只 fine-tune 高分辨率 decoder, 可以显著降低高分辨率训练成本; 较低 FPS 不是简单减少 I/O, 而是让模型看到更大的帧间变化, 改善 motion representation。[18]

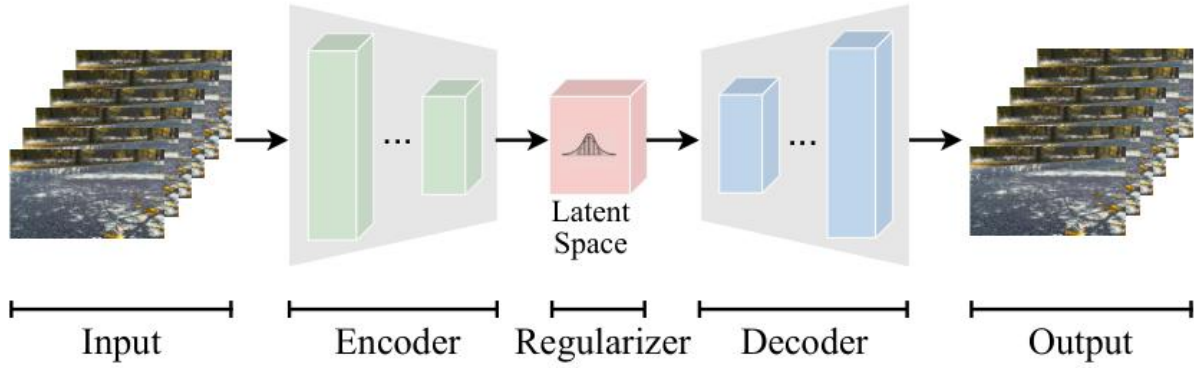


图4-8 VidTok 的模块化视频tokenizer 研究框架。

证据锚点 / Evidence anchor: VidTok 原文 Fig.2; text/ar5iv\_2412.13061.txt。

#### 4.6 LTX-Video: decoder 同时承担最后一步去噪

LTX-Video 把压缩提高到时间 8 倍、空间 32 倍，总 pixel-to-latent ratio 约 1:192。它把 patchification 前移进 VAE，并让 timestep-conditioned decoder 接收多层噪声注入，在 latent-to-pixel 重建时完成最后一步去噪。[21] 这打破了经典边界：decoder 不再只是固定 inverse transform，而成为生成轨迹的一部分。

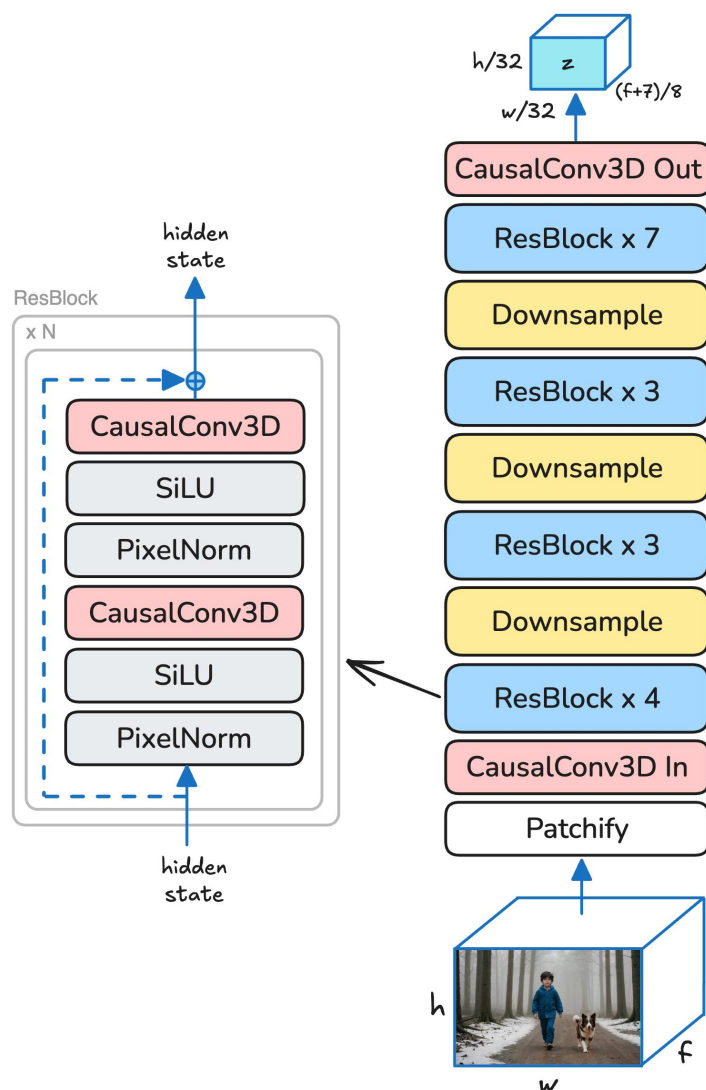


图4-9 LTX-Video VAE encoder 的高压缩路径。

证据锚点 / Evidence anchor: LTX-Video 原文 Fig.2 资产; text/ar5iv\_2501.00103.txt。

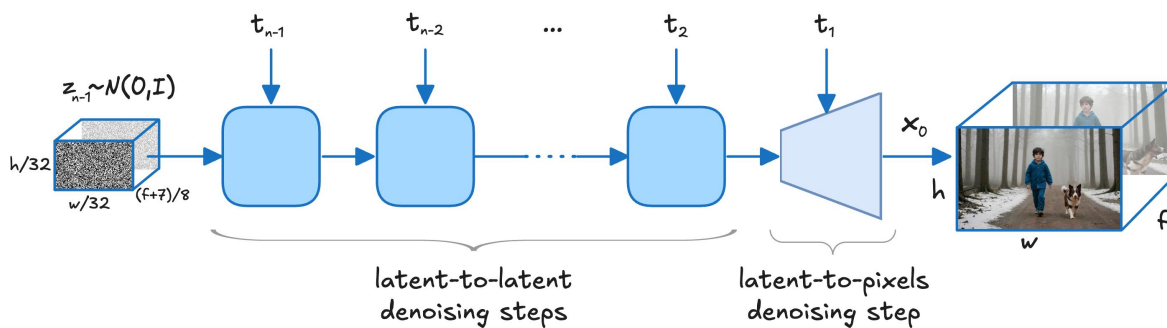


图4-10 LTX-Video 将最后一步 denoising 合入 decoder。

证据锚点 / Evidence anchor: LTX-Video 原文 Fig.4 资产。

**适用边界** | 这种 decoder 适合端到端实时生成，但与需要严格复用独立 VAE latent contract 的系统兼容性更弱。

#### 4.7 Cosmos：连续、离散与 diffusion decoder 共存

Cosmos tokenizer family 同时支持 continuous latent diffusion 与 discrete autoregressive modeling。其 encoder 使用 3D Haar wavelet、causal residual/downsampling 与 spatiotemporal attention；离散 token 还可通过 diffusion decoder 恢复到连续表征再生成视频。[22] 这说明未来 tokenizer 可能不再绑定单一生成范式。

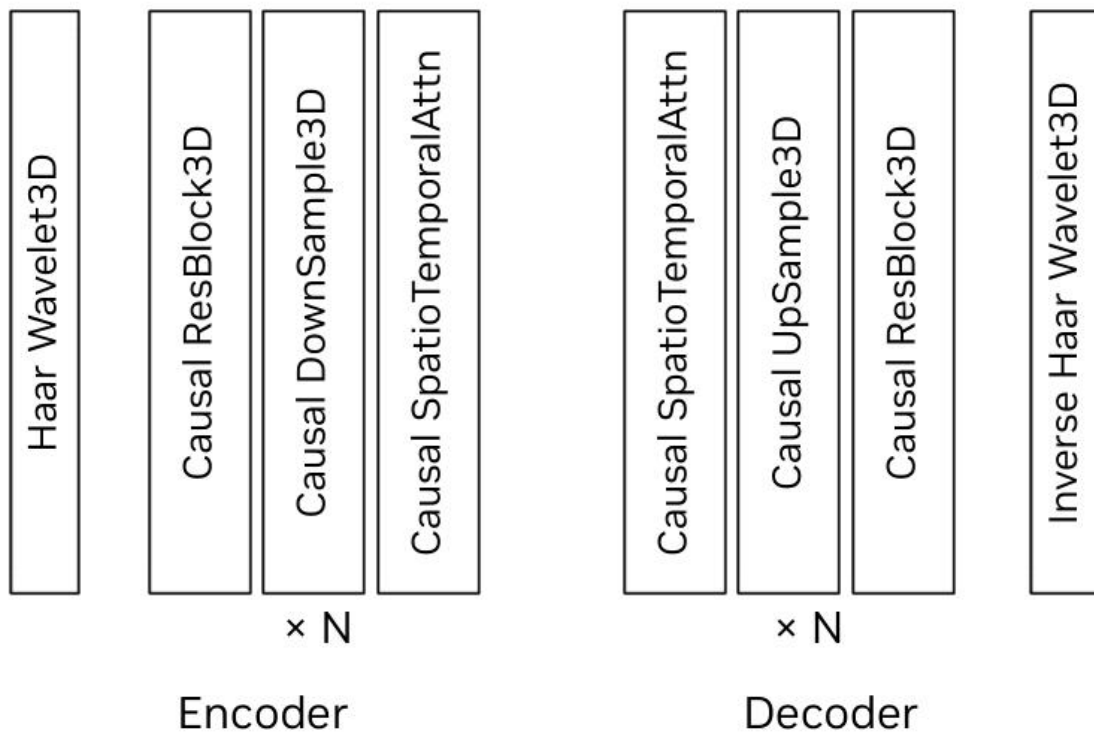


图 4-11 Cosmos tokenizer 的因果时空架构。

证据锚点 / Evidence anchor: Cosmos 原文网络架构图; text/ar5iv\_2501.03575.txt。



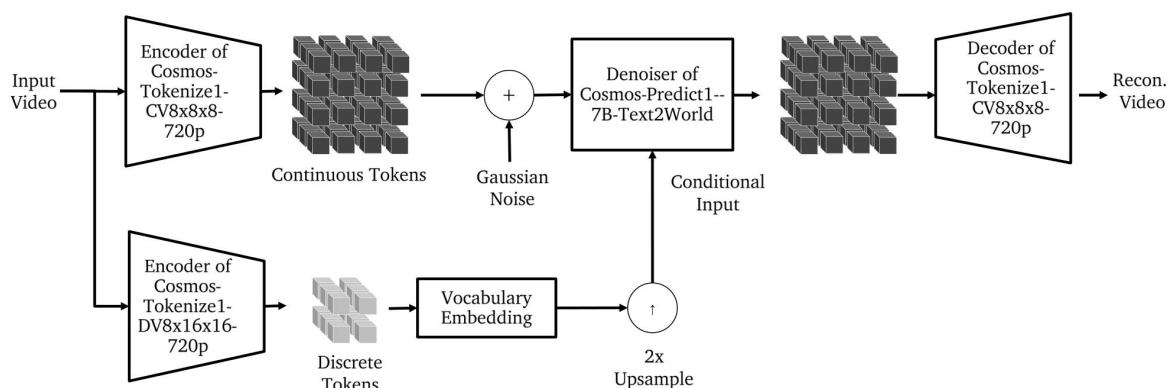


图4-12 Cosmos 的 discrete-to-continuous diffusion decoder。

证据锚点 / Evidence anchor: Cosmos 原文 diffusion decoder 图。

#### 4.8 Wan VAE: FlashVSR 所依赖的完整 causal contract

Wan VAE 使用 16-channel continuous causal latent，时间与空间压缩为 4x8x8。官方源码按首帧 1 帧、后续每 4 帧处理，并在 encoder/decoder 中维护 feature cache；mean/std 为 16 通道向量。

FlashVSR 的 WanDecoder 指的就是这条完整 decoder path。[23]

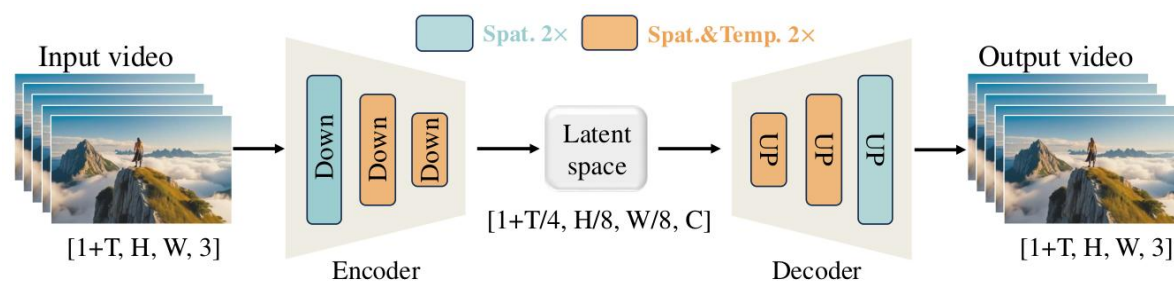


图4-13 Wan VAE 的 4x8x8 causal video latent 架构。

证据锚点 / Evidence anchor: Wan 原文 Fig.5 资产; source\_code/wan/wan/modules/vae.py:483-660。

实现层面，WanVAE\_ 同时实例化 Encoder3d 与 Decoder3d；encode 将视频拆成 1,4,4,... 帧块，decode 按 latent time step 逐块恢复，并在每次调用前后 clear\_cache。这个 cache contract 是流式正确性的组成部分，不能只复制卷积权重而忽略。

## 5. 编解码器与 DiT 的训练方法

### 5.1 三类训练问题必须分开

| 训练问题               | 输入/目标                                            | 更新参数                                              | 核心风险                  |
|--------------------|--------------------------------------------------|---------------------------------------------------|-----------------------|
| Tokenizer/VAE 预训练  | RGB/Video -> same RGB/Video                      | encoder + bottleneck + decoder, 可能含 discriminator | 压缩失真、GAN 幻觉、码本坍塌、时序闪烁 |
| 冻结 latent 的 DiT 训练 | latent + condition -> noise/velocity             | DiT/Flow Transformer                              | tokenizer 缺陷无法被主干修复   |
| 复原任务适配             | LQ observation + noisy latent -> HR latent/pixel | adapter/LoRA/condition projector/decoder          | 忠实度与感知生成冲突            |

### 5.2 Tokenizer/VAE 预训练的 LOSS 组合

$$L_{\text{tokenizer}} = \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{perc}} L_{\text{LPIPS}} + \lambda_{\text{reg}} L_{\text{KL/VQ}} + \lambda_{\text{adv}} L_{\text{GAN}} + \lambda_{\text{temp}} L_{\text{temp}} + \lambda_{\text{freq}} L_{\text{freq}}$$

通用分析式，不代表任一模型官方权重

- $L_{\text{rec}}$ : L1 通常比 L2 更少鼓励平均化; Charbonnier 可在异常值下更稳。严格复原优先保证这一项。
- $L_{\text{LPIPS}}$ : 改善语义和局部纹理，但会允许像素偏移。对人脸、文字、细线结构应配合 ROI 或 edge 约束。
- $L_{\text{KL}} / L_{\text{VQ}}$ : 决定 latent 规则性与可用容量。KL 太强导致信息丢失，VQ commitment 太弱导致 encoder 漂移。
- $L_{\text{GAN}}$ : 改善高频统计，但最容易引入伪纹理。通常在 reconstruction 稳定后延迟启用，或仅在高分辨率局部 patch 使用。
- $L_{\text{temp}}$ : 可由 optical-flow warp consistency、feature consistency、first/last-frame consistency 等组成；必须避免错误光流把运动边界抹平。
- $L_{\text{freq}}$ : 小波、FFT 或高频 band loss 可补充边缘，但不能替代像素与感知目标。WF-VAE 的 wavelet loss 是代表案例。

**推荐起点，不是定论** | 面向真实复原的连续 VAE 可先以 L1/Charbonnier 为主，加入较小 LPIPS 和 KL；GAN 延迟启用并从小权重开始。所有权重都应通过 gradient-norm logging、重建/感知双指标和消融实验校准，不能只复制论文常数。

### 5.3 冻结 latent 的 DiT / Flow Matching

经典 DiT 在冻结 VAE latent 上做 diffusion noise prediction；SD3/FLUX 路线使用 rectified flow / flow matching，学习从噪声到数据的 velocity field。encoder 和 decoder 通常不接收梯度，latent 可以预编码缓存，显著减少训练开销。[6][8]

$$x_t = (1-t) x_0 + t x_1 ; \quad v_{\text{target}} = x_1 - x_0 ; \quad L_{\text{FM}} = E \left[ \|v_{\text{theta}}(x_t, t, c) - v_{\text{target}}\|_2^2 \right]$$

线性概率路径上的 flow matching 简化式

预编码的好处是吞吐高、训练稳定；代价是 tokenizer 固定成为质量天花板。更换 VAE 后，即使空间尺寸相同，只要 latent channel、shift/scale 或统计分布不同，DiT 权重也不能直接复用。

### 5.4 高压缩和视频 VAE 为什么需要 curriculum

| 策略                                           | 代表模型      | 优化目的                    |
|----------------------------------------------|-----------|-------------------------|
| proxy code -> RGB                            | TiTok     | 先学 token 语义，再学高频像素      |
| 低分辨率全模型 -> 高分辨率<br>decoder-only              | VidTok    | 把高分辨率算力集中到解码器           |
| 低分辨率 -> HR latent adaptation -><br>local GAN | DC-AE     | 稳定 f64/f128 高压缩训练       |
| 2D feature target -> RGB video target        | Open-Sora | 从成熟图像表征逐步过渡到视频像素        |
| decoder 融合最后去噪                               | LTX-Video | 让 tokenizer 与最终生成轨迹共同优化 |

## 6. 数据集制作与模型输入

### 6.1 图像 tokenizer 数据

图像 VAE 的目标是覆盖将来生成或复原分布的视觉统计，而不是盲目追求海量 caption。基础流程包括：来源合法性检查、去重、质量筛选、尺寸和长宽比分桶、颜色空间统一、动态范围与 alpha 处理、随机 crop/resize，以及固定归一化到模型范围。生成数据集与 tokenizer 数据集可以不同，但 tokenizer 数据分布如果缺少文字、小脸、夜景或细纹理，decoder 会形成系统性盲区。

| 阶段 | 处理                                          | 必须记录的元数据                                         |
|----|---------------------------------------------|--------------------------------------------------|
| 入库 | 来源、许可、SHA/感知去重                              | source_id、license、resolution、color profile       |
| 质量 | 美学、清晰度、压缩、NSFW、异常图                          | quality scores、filter reason                     |
| 采样 | resolution/aspect buckets、global/local crop | crop box、resize kernel、bucket id                 |
| 输入 | RGB range、dtype、VAE encode                  | normalization、latent scale/shift、encoder version |

### 6.2 视频 tokenizer 数据

视频数据需要在图像清洗之外管理 shot boundary、帧率、运动量、duration、首帧策略和解码一致性。低 FPS 可以提高相邻训练帧的运动幅度，帮助模型学习 motion；但过低会产生非连续跳变。长视频通常应先切 shot，再按帧数或 token budget 分桶，并记录原始 FPS 与采样 FPS。

- 统一张量约定：推荐在数据层固定 [B,3,T,H,W]，模型内部若使用 [B,T,C,H,W] 必须显式转换并写单元测试。
- first-frame policy：首帧是否单独编码、是否重复 padding、后续按 4/8 帧块处理，必须进入数据与 cache contract。
- mixed-duration batching：按 token 数而非 clip 数控制 batch，避免长视频样本挤爆显存。CogVideoX Frame Pack 是代表方案。
- 时序清洗：检测重复帧、冻结帧、强转场、字幕条、极端抖动和可变帧率；这些问题会直接污染 causal decoder。

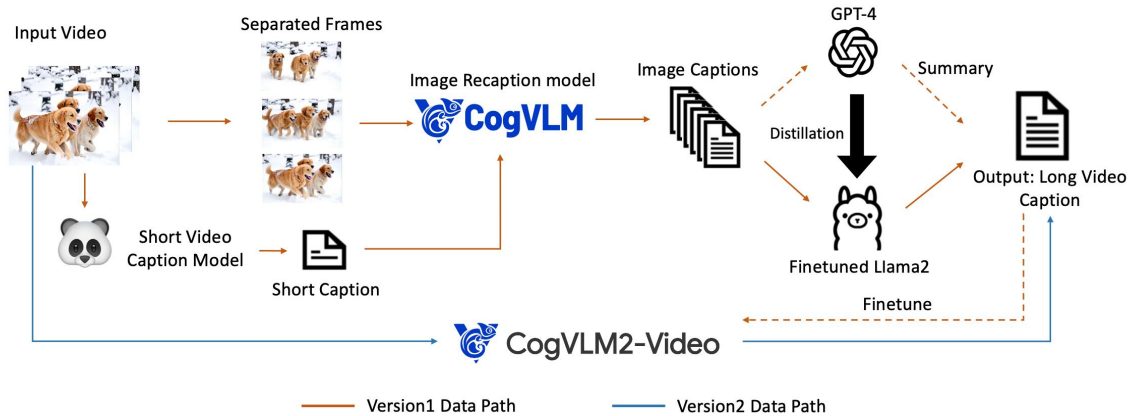


图6-1 CogVideoX 的质量与caption 数据处理管线。

证据锚点 / Evidence anchor: CogVideoX 原文 Fig.6-7 相关资产。

### 6.3 超分/复原配对数据

复原任务不能只靠 self-reconstruction。需要从 HQ/GT 生成或采集 LQ observation，并保留退化参数、对齐质量和任务 mask。真实 SR 的退化链应覆盖光学、传感器、ISP、resize、压缩和二次处理，而不是只有 bicubic。FlashVSR 使用 RealBasicVSR degradation 从 HQ 构造 LR-HR 配对。[24]

$$x\_LQ = \text{JPEG}(\text{Sharpen}(\text{Denoise}(\text{Resize}(\text{Blur}(x\_HQ)) + \text{Noise})))$$

示意退化链；实际顺序需随机化并记录参数

| 数据类型                   | 输入                              | 目标                     | 关键风险                             |
|------------------------|---------------------------------|------------------------|----------------------------------|
| 合成 paired SR           | 退化后的 LR/LQ                      | 原始 HQ                  | 退化分布过窄，模型只会识别脚本痕迹                |
| 真实 paired capture      | 多焦段/多曝光/多设备 LQ                  | 高质量参考帧                 | 亚像素、rolling shutter、曝光和 ISP 不一致  |
| unpaired real LQ       | 真实低质图像/视频                       | 无严格 GT                 | 需 teacher、cycle、GAN 或无参考奖励，幻觉风险高 |
| mask-aware restoration | LQ + face/text/edge/region mask | 区域化 HQ/identity target | 小区域梯度被全图平均，需要面积归一化               |

**数据审计** | 每个训练 sample 应至少能追溯 HQ source、LQ degradation seed/parameters、crop coordinates、frame indices、FPS、mask version、alignment score 和 tokenizer version。没有这些字



段，后续失败样本无法定位到数据还是模型。

## 6.4 可直接落地的输入检查

6. 检查 H、W 是否能被 VAE spatial factor 与 DiT patch size 整除；视频还要检查 T 与 first-frame/temporal factor 的兼容关系。
7. 把 latent mean/std、scale/shift、dtype、range 写入 checkpoint metadata；禁止靠代码默认值猜测。
8. encode-decode round trip 单独评测：PSNR/SSIM/LPIPS + face/text/edge 子集 + 视频 flicker/warp 指标。
9. 训练前可视化 64 个 batch：原图、crop、LQ、mask、VAE reconstruction、latent channel statistics。
10. 对 causal 模型做 prefix invariance test：截断未来帧后，过去输出应保持一致；否则 padding/normalization/cache 可能泄漏未来信息。

## 7. FlashVSR 深入案例：LQ\_proj\_in、WanDecoder 与 TCDecoder

FlashVSR 的关键贡献不是简单给 Wan 增加 SR condition，而是把实时流式 VSR 拆成三个可独立优化的系统瓶颈：条件输入、生成主干和像素解码。[24] 三个模块名称接近编解码概念，但边界完全不同。

## FlashVSR 三类编解码模块关系 / Module Boundary

LQ\_proj\_in 是条件投影, Wan Encoder/Decoder 是完整 VAE, TCD decoder 是 decoder-only 加速器。

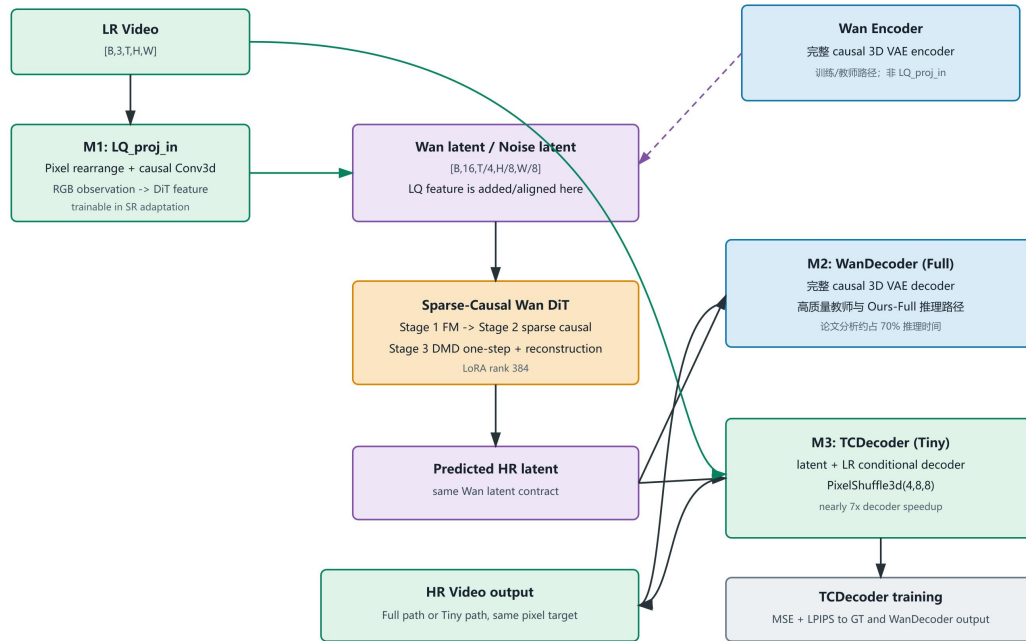


图 7-1 FlashVSR 三类模块的边界与数据流。

证据锚点 / Evidence anchor: 解释图; 依据 FlashVSR 原论文、utils.py、TCD decoder.py 和 Wan VAE 源码绘制。

| 模块          | 真实类别                            | 输入 -> 输出                                                                       | 推理职责                                                                           |
|-------------|---------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| LQ_proj_in  | conditional projection          | LR RGB video [B,3,T,H,W] -> LQ condition feature aligned with noisy Wan latent | replaces VAE encoder for LR conditioning                                       |
| Wan Encoder | full causal VAE encoder         | HR video [B,3,T,H,W] during VAE/teacher paths -> 16-channel Wan latent         | not required for LR condition injection in the final FlashVSR path             |
| WanDecoder  | full causal VAE decoder         | 16-channel Wan latent -> RGB HR video                                          | FlashVSR Full quality path; about 70% of runtime at 768x1408 in paper analysis |
| TCD decoder | tiny causal conditional decoder | predicted Wan latent plus LR RGB conditional features -> RGB HR video          | Ours-Tiny path; nearly 7x decoder speedup in paper                             |

## 7.1 VSR-120K 与三阶段训练

FlashVSR 从 Videvo、Pexels、Pixabay 收集初始 600K 视频和 220K 图像, 使用 LAION-Aesthetic、MUSIQ 和 RAFT 做质量与运动过滤, 最终得到 120K 视频 (平均超过 350 帧) 和 180K HQ 图像。LR-HR 配对由 RealBasicVSR degradation 合成。Stage 1 把图像视为 T=1 视频联合训练; Stage 2 转为 block-sparse causal attention; Stage 3 用 DMD 压到一步生成。

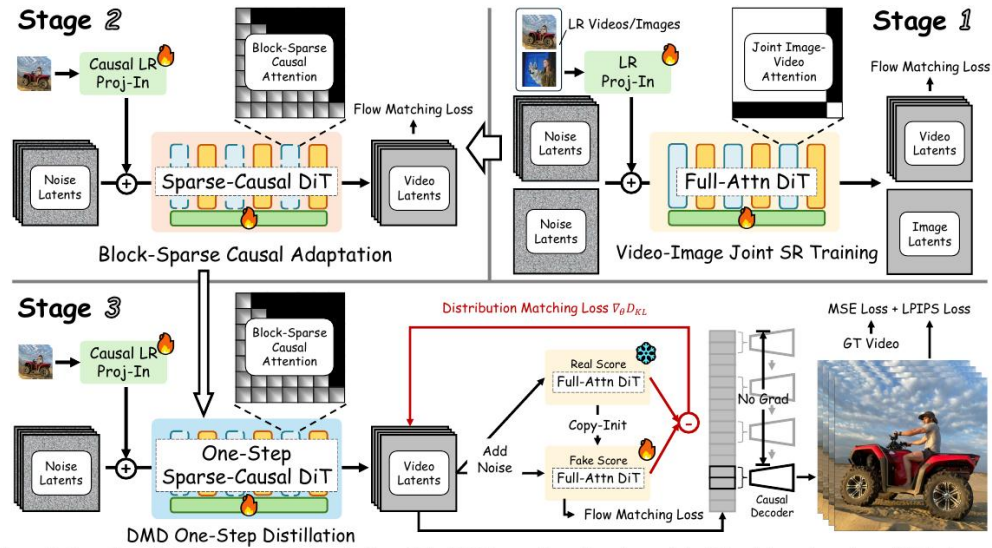


Figure 2. Overview of the three-stage training pipeline of FlashVSR, covering video-image joint SR training, adaptation with block-sparse causal attention for streaming inference, and distribution-matching one-step distillation combined with reconstruction supervision.

**Stage 1. Video-Image Joint SR Training.** We adapt a pre-trained video diffusion model (WAN2.1 1.3B [38]) to super-

level features to compute a coarse block-to-block attention map. The top- $k$  most relevant block pairs are selected, and

图 7-2 FlashVSR 的数据、条件投影与三阶段训练流程。

证据锚点 / Evidence anchor: FlashVSR 原论文第 4 页 Fig.2; 本地 PDF: papers/05\_flashvsr\_case/flashvsr\_2510.12747.pdf。

$$L_{\text{stage3}} = L_{\text{DMD}} + \lambda_{\text{FM}} L_{\text{FM}} + \lambda_{\text{MSE}} L_{\text{MSE}} + 2 * L_{\text{LPIPS}}$$

论文公开的 Stage 3 组合; 其他系数按原文/实现核验

实现设置包括 LoRA rank 384、AdamW、learning rate 1e-5、weight decay 0.01、batch size 32。这里训练的是 restoration DiT/adaptor，不是从头训练 Wan VAE。

## 7.2 LQ\_proj\_in: 条件投影，不是完整 encoder

源码中的 Buffer\_LQ4x\_Proj 先用 PixelShuffle3d-style rearrangement 把 16x16 空间块搬到 channel，再用两个 causal Conv3d 将时间连续下采样两次，并通过 RMSNorm、SiLU 与逐层 Linear 映射到 DiT feature space。forward 还复制首帧并按 4 帧块处理，以对齐 Wan causal latent。

```
[B,3,T,H,W] -> rearrange(1,16,16) -> causal Conv3d x2 -> per-block DiT condition features
```

source\_code/flashvsr/utils.py:54-105

它替代的是 LR condition 的 VAE encoding 路径，而不是生成一个可由标准 Wan decoder 独立解码的 posterior。它没有 mu/logvar、reparameterization，也不是一个完整 tokenizer。训练监督来自 restoration flow objectives 间接传递。

**常见误读** | 把 LQ\_proj\_in 称为 LQ encoder 会掩盖它与 Wan latent posterior 的统计差异。更准确的名称是 latent-aligned condition projector。

### 7.3 WanDecoder：完整、高质量但昂贵的 causal VAE decoder

WanDecoder 接收 16-channel Wan latent，经 conv2、Decoder3d、时空上采样与 feature cache 恢复 RGB 视频。官方源码的 decode 按 latent time step 迭代，并在每一步复用 causal feature cache。FlashVSR 论文分析中，完整 Wan decoder 在 768x1408 设置下约占 70% 推理时间，因此即使 DiT 已经一步化，decoder 仍是系统瓶颈。

**证据锚点 / Evidence anchor:** *source\_code/wan/wan/modules/vae.py:483-660; source\_code/flashvsr/wan\_video\_vae.py; FlashVSR p5 Sec.3.4。*

### 7.4 TCDecoder：decoder-only 条件蒸馏器

TCDecoder 在源码中是 TAEHV decoder-only：输入 predicted Wan latent，使用 MemBlock 保存前帧状态，经过三次空间上采样、两次时间增长，最后输出 RGB；PixelShuffle3d(4,8,8) 负责把 Wan 压缩尺度重排到 decoder 可消费的时空结构。它还接收 LR conditional features，以减少小 decoder 完全依赖 latent 猜测细节的压力。

| 源码模块             | 锚点                   | 作用与训练含义                                                                           |
|------------------|----------------------|-----------------------------------------------------------------------------------|
| PixelShuffle3d   | TCDecoder.py:73-89   | 按 (4,8,8) 重排时空压缩维度，建立 Wan latent 到轻量 decoder 的输入接口                                |
| TAEHV            | TCDecoder.py:170-213 | decoder-only 主体；组合条件特征、残差块与多级上采样并输出 RGB                                           |
| MemBlock         | causal state         | 保存前帧特征状态，维持流式解码中的因果一致性与跨帧连续性                                                      |
| TGrow            | temporal upsampling  | 逐级增长时间维，恢复 Wan 4 倍时间压缩对应的帧序列                                                      |
| Dual supervision | GT + Wan teacher     | 对 GT 和 WanDecoder output 分别计算 MSE + LPIPS，兼顾真实目标与 latent-compatible teacher prior |

venting artifacts. Right: Two local window rules, namely boundary-preserved attention mask is computed within these local masks.

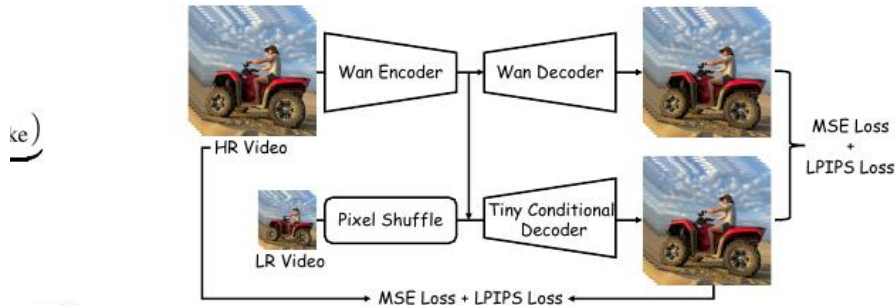


Figure 4. Training pipeline of the TC Decoder.

*spatial neighborhood*, thereby aligning the effective attention range with that used during training. With the relative formulation of RoPE [35], this simple constraint eliminates the train–inference mismatch in positional range. As shown in the middle of Fig. 3, this approach closes the resolution gap and enables consistent performance even on ultra-high-resolution outputs, *without requiring any spatial tiling*. This design is further validated through quantitative results in Sec. 4.4 and qualitative visualizations in Appendix.

### 3.4. Tiny Conditional Decoder

图 7-3 FlashVSR 原论文的 TCDecoder 条件解码与蒸馏路径。

证据锚点 / Evidence anchor: FlashVSR 原论文第 5 页 Fig.4; source\_code/flashvsr/TCDecoder.py:73-213。

$$L_{TC} = (L_{MSE}(y_{tiny}, y_{GT}) + L_{LPIPS}(y_{tiny}, y_{GT})) + (L_{MSE}(y_{tiny}, y_{Wan}) + L_{LPIPS}(y_{tiny}, y_{Wan}))$$

同时监督 GT 与 WanDecoder teacher

双目标设计的意义是互补：GT 监督纠正 teacher 的系统偏差，Wan decoder output 则提供更平滑、与 latent contract 一致的蒸馏目标。论文在 61-frame 384x384 clips 上单独训练约两天，并报告 decoder 接近 7 倍加速。这个速度是 decoder 子系统的论文结果，不应误写成整个端到端系统 7 倍。



## 7.5 Full 与 Tiny 的质量-速度边界

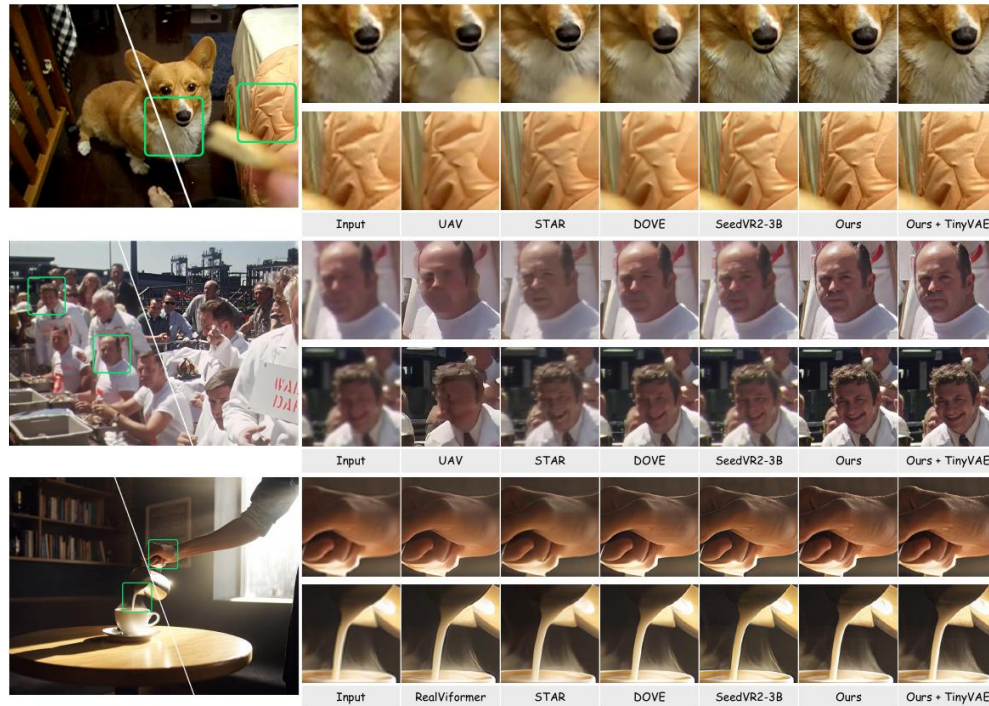


Figure 5. Visualization results of video super-resolution on real-world and AIGC videos.

Table 3. 13.6% sparse attention achieves similar reconstruction quality to full attention with  $\sim 7.4\times$  lower computation.

图 7-4 FlashVSR 原文定性结果：流式超分的细节与时序表现。

证据锚点 / Evidence anchor: FlashVSR 原论文第 7 页定性对比。

Full path 保留完整 Wan decoder，适合离线高质量；Tiny path 用 TCD decoder 换取实时性。产品上应把两者视为同一 latent generator 的两个 pixel renderer，而不是两个独立 SR 模型。切换 decoder 时要分别评测：纹理真实感、文字/人脸忠实度、运动边界、首帧冷启动、cache reset 和长视频漂移。

**对 FlashVSR 的系统洞察** | 一步 DiT 并不等于实时系统。真正的端到端 latency 由 condition projection、attention、latent update、decoder、I/O 和 cache 管理共同决定；FlashVSR 的 TCD decoder 证明 decoder-only 替换可以成为独立的加速方向。



## 8. 架构选型与复现建议

### 8.1 按任务选择 latent contract

| 任务       | 优先选择                                                 | 谨慎选择                 | 理由                         |
|----------|------------------------------------------------------|----------------------|----------------------------|
| 高保真图像复原  | f8 continuous VAE; 4-16ch; 弱 GAN                     | f64/f128 高压缩         | 需要保留文字、人脸和细线结构             |
| 高分辨率图像生成 | SD3/FLUX 16ch f8; DC-AE f32/f64                      | 极低容量离散 token         | 平衡 token 数与 decoder 细节     |
| 长视频生成    | causal 3D VAE + first-frame/cache contract           | 非因果 3D tokenizer     | 流式、分块和长时一致性                |
| 实时视频超分   | Wan latent + 条件 projector + tiny conditional decoder | 完整 VAE encoder 处理 LR | 条件输入和 pixel renderer 都需轻量化 |
| 端侧部署     | 低通道/低 token、decoder-only distillation、量化友好卷积         | 复杂 attention decoder | 内存、启动和功耗比单次 FLOPs 更关键      |

### 8.2 评测不应只看重建 PSNR

- Tokenizer 层：PSNR/SSIM、LPIPS/DISTS、rFID、face identity、OCR/line fidelity、color delta、temporal flicker。
- 生成系统层：FID/FVD、prompt alignment、motion quality、sampling steps、peak VRAM、encode/decode latency。
- 复原层：paired fidelity、no-reference quality、hallucination audit、区域化指标、真实相机数据盲测。
- 流式层：first-frame latency、steady-state FPS、cache memory、chunk boundary consistency、随机 seek/reset 行为。
- 兼容层：latent scale/shift、channel order、first-frame policy、padding、normalization、checkpoint metadata。

### 8.3 最小可复现检查表

11. 固定 tokenizer checkpoint、commit SHA、latent channel、spatial/temporal factor、scale/shift 和输入 range。
12. 先跑 encode-decode round trip，确认重建与官方样例一致，再训练 DiT。
13. 单独记录 tokenizer 与 DiT 的数据清单，避免把 caption 数据误当 VAE 训练数据。
14. 所有 LOSS 权重输出 gradient norm；至少做 rec-only、+perceptual、+GAN、+temporal/frequency 的消融。
15. 视频模型测试不同 T：1、2、4、5、17、61 帧，覆盖 first-frame 和非整除边界。
16. decoder 替换时冻结 latent 输入，比较完整 decoder 与轻量 decoder 的逐样本差异。
17. 所有速度数据写明硬件、dtype、分辨率、帧数、batch、warm-up、是否含 VAE 和 I/O。

## 9. 综合洞察与研究空白

**Tokenizer 正在从预处理变成主架构变量** | TiTok、DC-AE、LTX-Video 和 Cosmos 都在改变 token 数、拓扑或 decoder 职责。未来论文需要把 tokenizer 与 DiT 联合报告，而不是只写 backbone 参数量。

**高压缩与严格恢复存在结构性冲突** | 越强的压缩越依赖 decoder 生成缺失高频。生成任务可接受 plausibility，真实相机复原却需要 evidence-grounded fidelity，因此应提供可控感知档位和幻觉审计。

**视频因果性是一份完整协议** | causal convolution 不足以保证流式正确；padding、normalization、first-frame、cache reset、chunk overlap 都必须一致。

**Decoder 是独立的性能优化对象** | FlashVSR 与 LTX-Video 从不同方向说明 decoder 可以承担条件恢复、最后去噪或独立蒸馏。模型加速不能只盯 Transformer。

**公开结构不等于公开训练** | FLUX.1 和 Wan 的架构可由官方源码核验，但完整 AE/VAE 数据与损失仍未披露。复现报告必须把 architecture reproduction 与 recipe reproduction 分开。

**数据工程决定可学到的边界** | 分辨率 curriculum、FPS、motion filtering、first-frame 和退化链不是外围实现，而是模型方法的一部分。

## 9.1 值得继续研究的问题

- 面向真实相机复原的可逆或近可逆高压缩 tokenizer，能否同时保留 RAW/ISP 证据与生成先验？
- 能否用 mask/uncertainty map 控制 decoder 只在证据不足区域生成细节，在可靠区域保持像素忠实？
- 视频 decoder distillation 如何避免长时漂移，并把 teacher 的 cache behavior 一并蒸馏？
- 是否可以为同一 latent generator 训练多个 renderer：faithful、aesthetic、mobile、cloud，并共享可校准的质量尺度？
- encoder/decoder 的量化误差如何与 flow trajectory、latent normalization 和一步蒸馏共同建模？
- 公开 benchmark 应如何同时测量重建、生成、因果流式和端侧成本，而不是单一 rFID/FVD？

## 附录 A：架构总表

以下为正文使用的紧凑对照。完整字段、证据锚点和备注请查阅同目录 Excel 的“架构比较”Sheet。

| 模型                   | 类别                                              | 压缩                                                                             | 通道/Token                                                                        | 证据                           |
|----------------------|-------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------|
| VAE                  | image continuous latent                         | task-dependent; no fixed modern f8 contract                                    | task-dependent                                                                  | primary-source-indexed       |
| VQ-VAE               | image discrete tokenizer                        | configuration-dependent                                                        | discrete code indices                                                           | primary-source-indexed       |
| VQGAN                | image discrete tokenizer                        | typically f16 in the paper's high-resolution tokenizer settings                | 2D code grid                                                                    | paper-verified               |
| LDM / SD-VAE         | image continuous KL autoencoder                 | f4-f32 studied; f8 became the common production point                          | commonly 4 channels at f8                                                       | paper-verified+code-verified |
| DiT latent interface | frozen image VAE plus transformer               | f8 VAE; DiT adds latent patch size p                                           | 4-channel latent in original DiT setup                                          | paper-verified+code-verified |
| SD3 VAE              | image continuous KL autoencoder                 | f8                                                                             | 16 channels                                                                     | paper-verified+code-verified |
| FLUX.1 AE            | image continuous autoencoder                    | f8 inferred from four resolution levels                                        | 16 channels                                                                     | code-verified                |
| TiTok                | image discrete 1D tokenizer                     | 32-256 tokens; 32-token flagship at 256px                                      | K tokens; codebook 4096 in final setting                                        | paper-verified               |
| DC-AE                | image continuous high-compression autoencoder   | up to f128; f64 emphasized for DiT acceleration                                | configuration-dependent continuous latent                                       | paper-verified               |
| MAGViT               | video discrete tokenizer                        | architecture/configuration-dependent                                           | 3D discrete token grid                                                          | paper-verified               |
| MAGViT-v2            | video/image discrete tokenizer                  | configuration-dependent                                                        | binary-factorized discrete codes                                                | paper-verified               |
| CogVideoX 3D VAE     | video continuous causal VAE                     | $T \times H \times W = 4 \times 8 \times 8$                                    | paper studies 4/8/16/32 channels; release config must be checked per checkpoint | paper-verified+code-verified |
| Open-Sora 1.2 VAE    | hybrid image plus video autoencoder             | $T \times H \times W = 4 \times 8 \times 8$                                    | not fully specified in report                                                   | paper-verified               |
| WF-VAE               | video continuous causal VAE                     | model variants evaluated at common video VAE ratios                            | configuration-dependent                                                         | paper-verified               |
| VidTok               | video continuous or discrete tokenizer          | multiple settings under a unified benchmark                                    | configuration-dependent                                                         | paper-verified               |
| LTX-Video VAE        | video continuous causal denoising VAE           | $T \times H \times W = 8 \times 32 \times 32$ ; pixel-to-latent ratio 1:192    | high-dimensional continuous latent                                              | paper-verified               |
| Cosmos Tokenizer     | video continuous and discrete causal tokenizers | continuous $4/8 \times 8 \times 8$ ; discrete $8 \times 16 \times 16$ variants | variant-dependent                                                               | paper-verified               |
| Wan VAE / WanDecoder | video continuous causal VAE                     | $T \times H \times W = 4 \times 8 \times 8$                                    | 16 channels                                                                     | paper-verified+code-verified |
| FlashVSR LQ_proj_in  | restoration condition projector                 | spatial rearrangement plus temporal causal projection                          | inference model-specific                                                        | paper-verified+code-verified |
| FlashVSR TCDecoder   | restoration decoder-only adapter                | inverts Wan $4 \times 8 \times 8$ scale using PixelShuffle3d(4,8,8)            | latent channels follow Wan; output RGB                                          | paper-verified+code-verified |

附录 B：LOSS 与训练阶段总表

| 模型                      | 训练范式                                                   | 损失                                                                                                                            | 边界                                                                                          |
|-------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| VAE                     | tokenizer pretraining                                  | negative ELBO = reconstruction likelihood term + KL divergence                                                                | Modern perceptual VAEs usually add objectives beyond the original VAE.                      |
| VQ-VAE                  | tokenizer pretraining                                  | reconstruction + codebook + commitment; straight-through estimator                                                            | Codebook collapse and dead codes require monitoring.                                        |
| VQGAN / LDM autoencoder | perceptual tokenizer pretraining                       | pixel reconstruction + LPIPS/perceptual + adversarial + codebook or KL regularization                                         | GAN weight must be tuned after reconstruction is stable.                                    |
| DiT                     | frozen-latent diffusion                                | diffusion noise prediction and learned variance objective                                                                     | The decoder is a reconstruction ceiling but receives no DiT gradient.                       |
| SD3 / MM-DiT            | frozen-latent rectified flow                           | conditional flow matching / rectified-flow objective                                                                          | The complete VAE pretraining recipe is not disclosed in this paper.                         |
| FLUX.1 AE               | tokenizer pretraining then frozen-latent flow training | undisclosed for AE pretraining                                                                                                | Do not copy SD3 or LDM loss weights and label them as FLUX facts.                           |
| TiTok                   | two-stage tokenizer training                           | stage 1 proxy-code prediction; stage 2 standard VQGAN pixel/perceptual/adversarial recipe                                     | Proxy codes simplify unstable direct RGB/GAN optimization.                                  |
| DC-AE                   | decoupled high-resolution autoencoder training         | reconstruction/perceptual first; high-resolution latent adaptation; lightweight local GAN refinement                          | Full-resolution GAN training is intentionally avoided for cost and stability.               |
| Open-Sora 1.2 VAE       | three-stage video VAE bootstrapping                    | stage 1 reconstruct 2D-VAE features + identity alignment; stage 2 feature reconstruction; stage 3 direct video reconstruction | An example of curriculum in target space, not merely resolution curriculum.                 |
| WF-VAE                  | video VAE pretraining                                  | L1 reconstruction + LPIPS/adversarial/KL family + wavelet loss; reported lambda_WL=0.1                                        | Paper reports L1 outperforming L2 in its setting.                                           |
| VidTok                  | two-stage resolution curriculum                        | continuous or discrete tokenizer objectives plus reconstruction/perceptual components                                         | Exact objective differs between continuous and discrete variants.                           |
| LTX-Video VAE           | joint denoising-aware video VAE training               | shared diffusion objective + Reconstruction GAN + reconstruction/perceptual family                                            | This deliberately blurs the classic boundary between diffusion last step and decoder.       |
| Wan VAE                 | video VAE pretraining                                  | complete public loss composition is not fully disclosed in the system report                                                  | Architecture is auditable; training weights and full data recipe are not.                   |
| FlashVSR stages 1-3     | restoration adaptation and one-step distillation       | stage 1 flow matching; stage 2 flow matching; stage 3 DMD + flow matching + MSE + 2*LPIPS                                     | This trains the restoration DiT, not the base Wan VAE.                                      |
| FlashVSR TCDecoder      | decoder-only conditional distillation                  | MSE+LPIPS to GT plus MSE+LPIPS to Wan decoder output                                                                          | Teacher reconstruction stabilizes detail while direct GT supervision corrects teacher bias. |

附录 C：术语中英对照

| English   | 中文     | Definition                                                                               | Pitfall                                                                       |
|-----------|--------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Encoder   | 编码器    | Maps pixels or video frames into a latent representation.                                | A condition projector may resemble an encoder but need not model a posterior. |
| Decoder   | 解码器    | Maps latent representations back to pixels or frames.                                    | A denoising decoder may also perform a diffusion step, as in LTX-Video.       |
| Tokenizer | 视觉分词器  | An encoder-bottleneck-decoder system defining the tokens consumed by a generative model. | Tokens may be continuous tensors or discrete indices.                         |
| VAE       | 变分自编码器 | A probabilistic autoencoder with a regularized latent distribution.                      | Modern 'VAE' implementations may use perceptual and adversarial objectives    |



| English                    | 中文      | Definition                                                                    | Pitfall                                                                                                   |
|----------------------------|---------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| VQ                         | 向量量化    | Maps continuous encoder outputs to discrete codebook entries.                 | beyond the original ELBO. Codebook collapse and inactive entries can reduce effective capacity.           |
| Latent channel             | 潜变量通道   | Feature dimension at each latent spatial-temporal position.                   | More channels improve reconstruction but make the DiT prediction problem harder.                          |
| Spatial compression factor | 空间压缩倍率  | Ratio from H,W pixels to latent H/f,W/f.                                      | It is distinct from DiT patch size and from total token compression.                                      |
| Temporal causality         | 时间因果性   | Output at time t does not depend on future frames.                            | A nominal causal convolution can still break streaming if normalization or padding uses future context.   |
| Perceptual loss            | 感知损失    | Feature-space distance, commonly LPIPS, used to preserve perceptual detail.   | It can trade pixel fidelity for plausible texture and must be balanced with reconstruction losses.        |
| Adversarial loss           | 对抗损失    | Discriminator-based objective encouraging realistic local statistics.         | Starting too early or weighting too highly can destabilize reconstruction.                                |
| Frozen latent training     | 冻结潜空间训练 | Precompute or freeze tokenizer outputs while training the DiT.                | No gradient improves decoder defects; tokenizer quality is a hard ceiling.                                |
| Conditional decoder        | 条件解码器   | Decoder that consumes latent plus an auxiliary observation such as LR frames. | It can recover details from the observation but may over-copy degradations without robust training pairs. |

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